



LECTURE 6:

ATTENTION AND TRANSFORMER

University of Washington, Seattle

Fall 2025



OUTLINE

Part 1: Transformer motivation

- Limitation of RNNs with sequence data
- Seq2seq and attention
- Attention is all you need

Part 2: Transformer architecture

- Encoder
- Decoder
- Transformer

Part 3: Encoder architecture

- Positional encoding
- Self-attention layer
- Residual connection
- Layer normalization

Part 4: Decoder architecture

- Masked multi-headed attention

Part 5: Transformer applications

- Encoder only vs Decoder only



Transformer Motivation

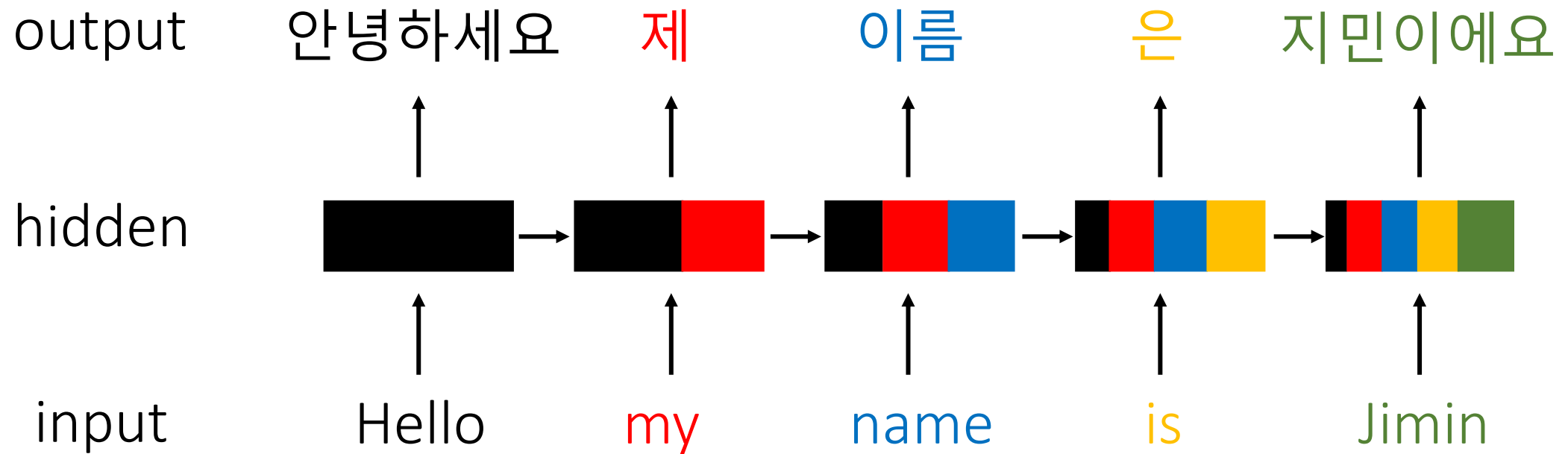
Limitations of RNNs with sequence data

Seq2Seq and attention

Attention is all you need



Limitations of RNNs



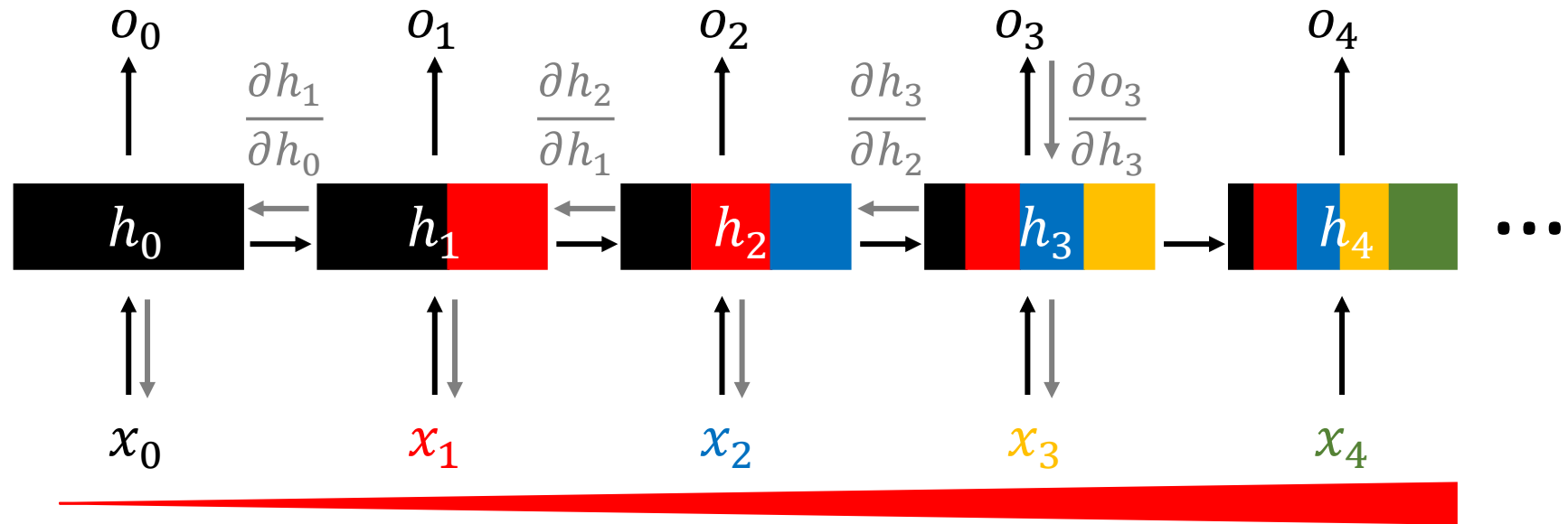
Vanishing and Exploding Gradients

→ Forward
← Backward

output

hidden

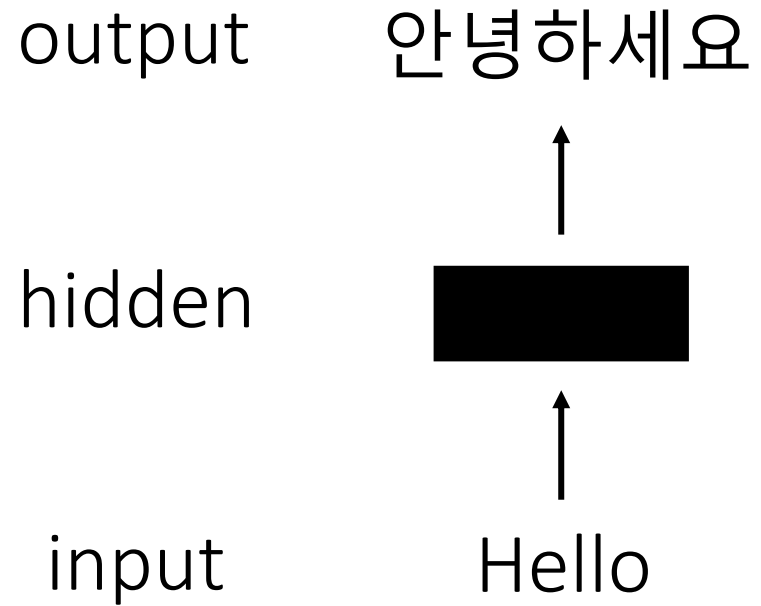
input



Longer input sequence →
higher risk of Vanishing/Exploding Gradients!

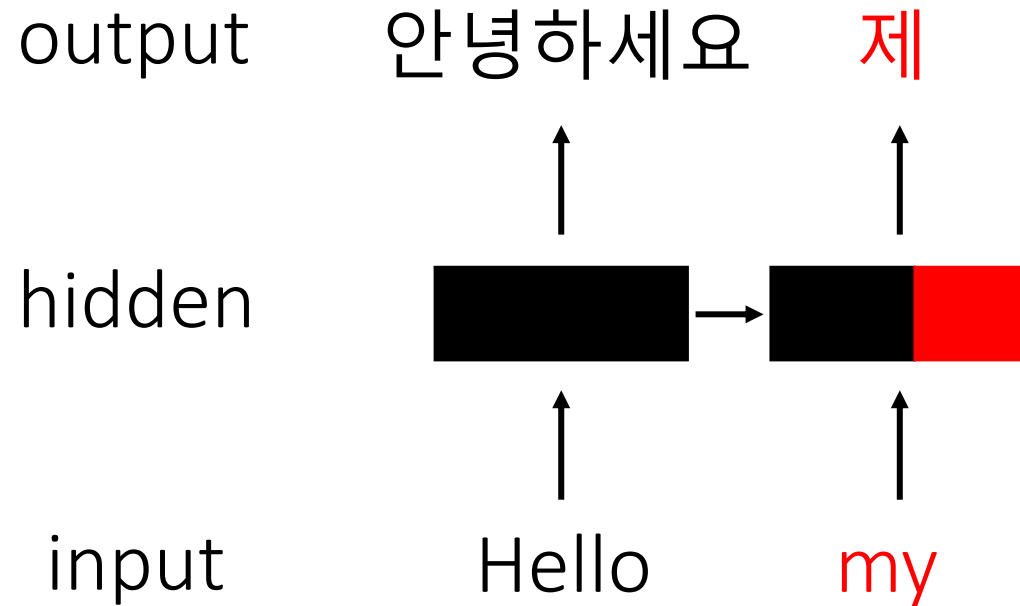


RNN Architecture



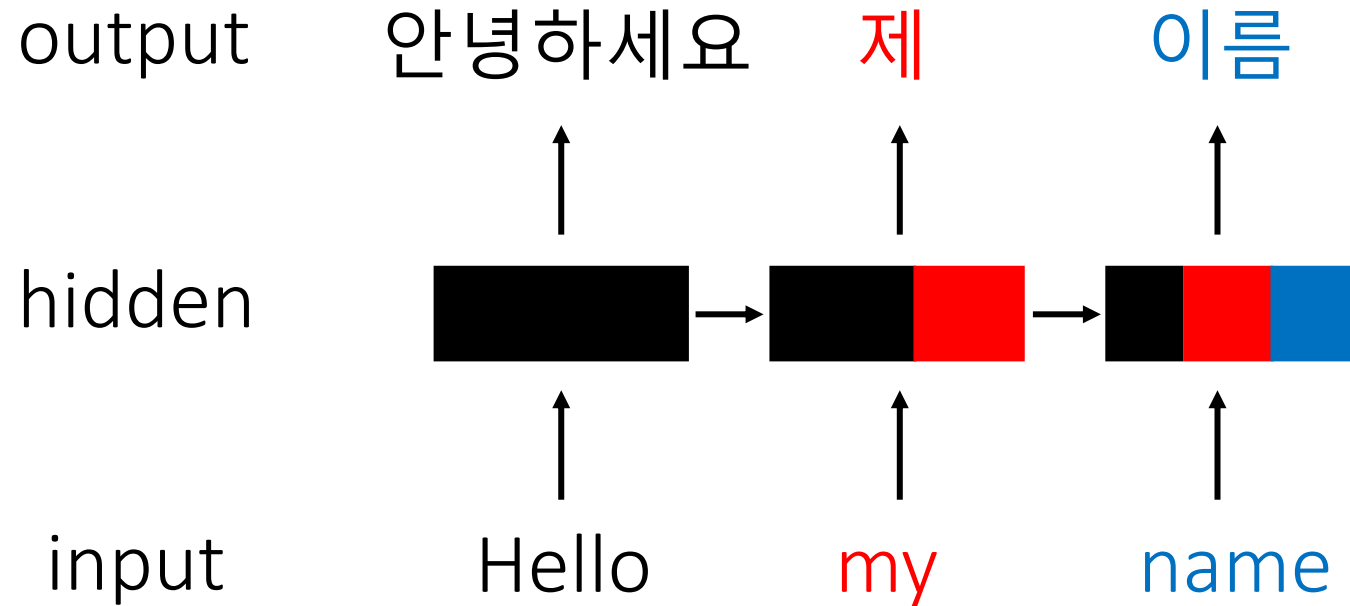


RNN Architecture



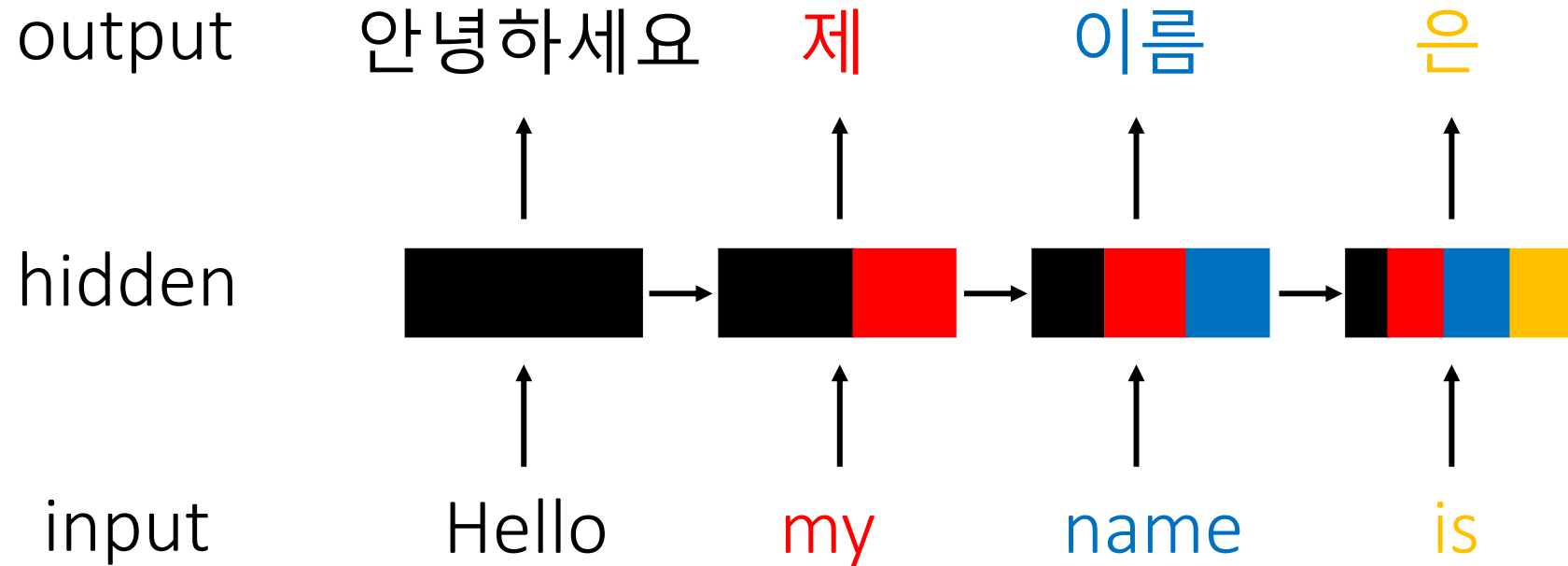


RNN Architecture



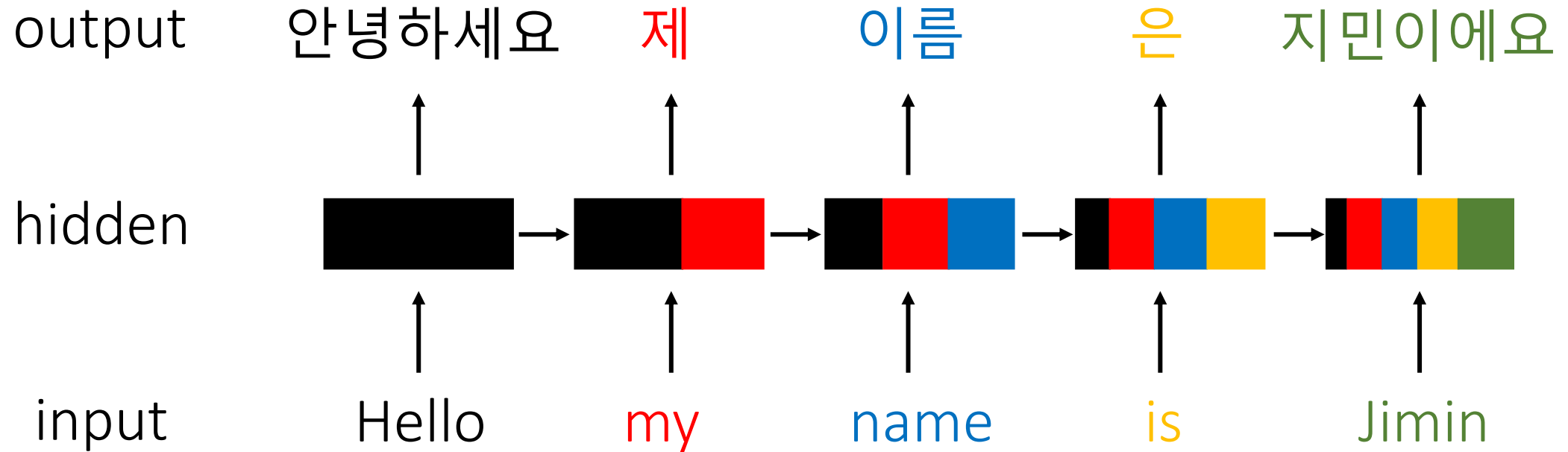


RNN Architecture



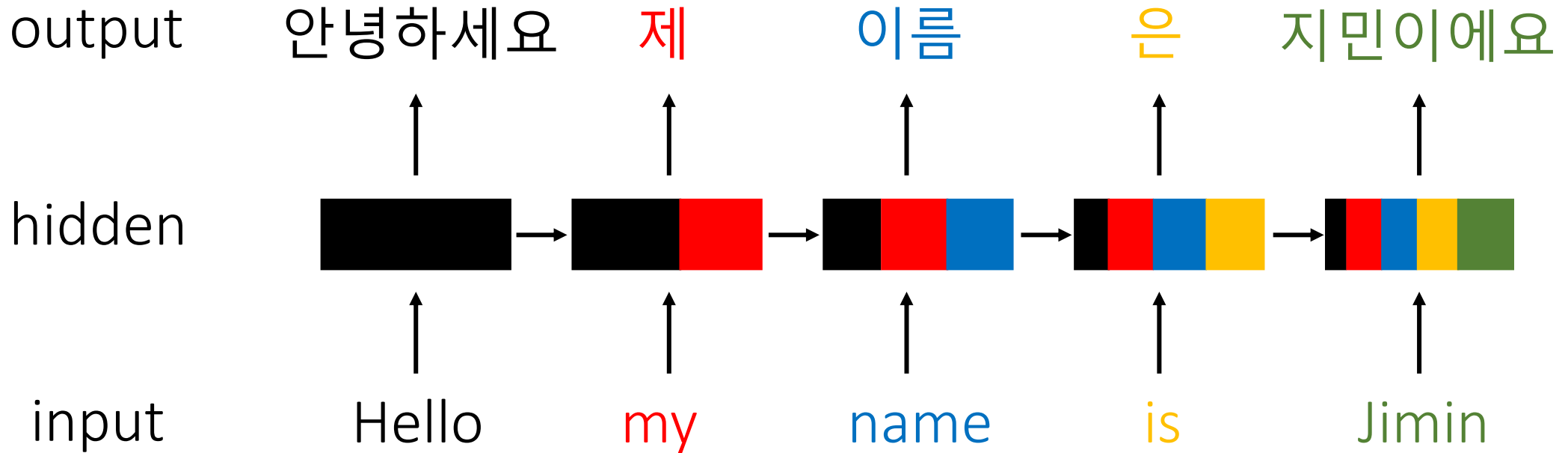


RNN Architecture





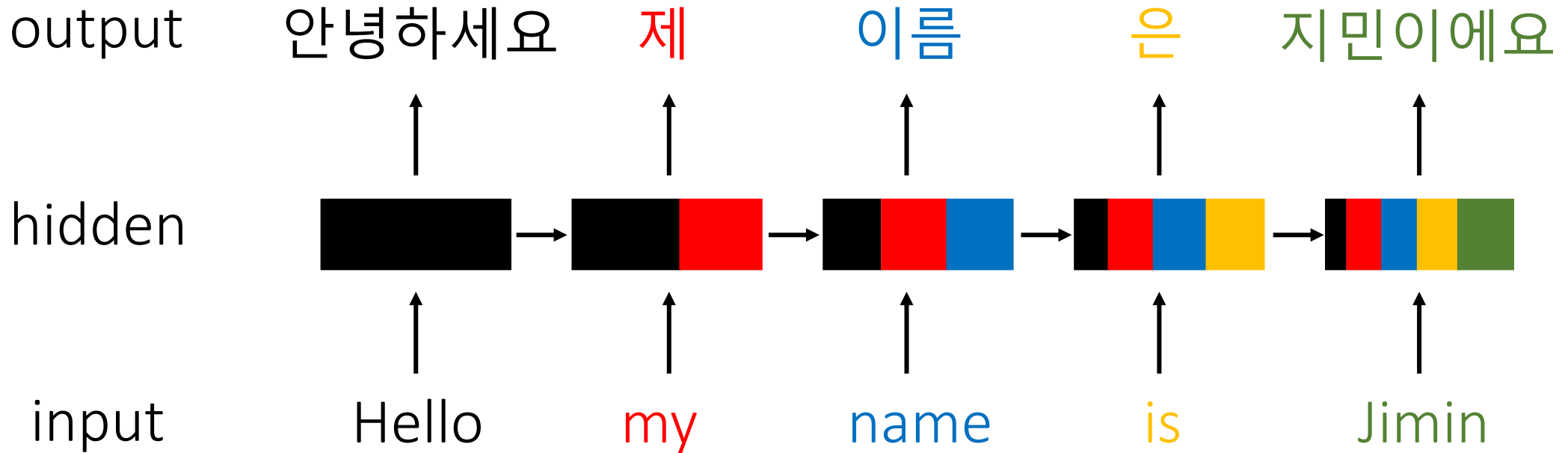
RNN Architecture



Each input (token) is fed sequentially →
No parallelization



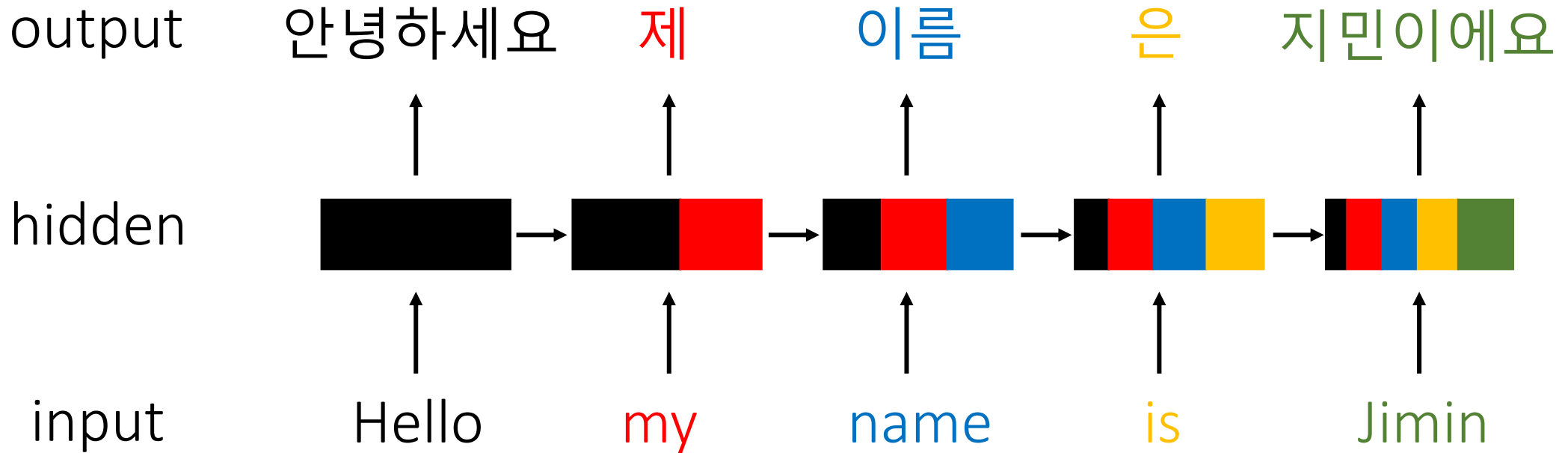
RNN Architecture



Difficult to store long-term context when sequence is long



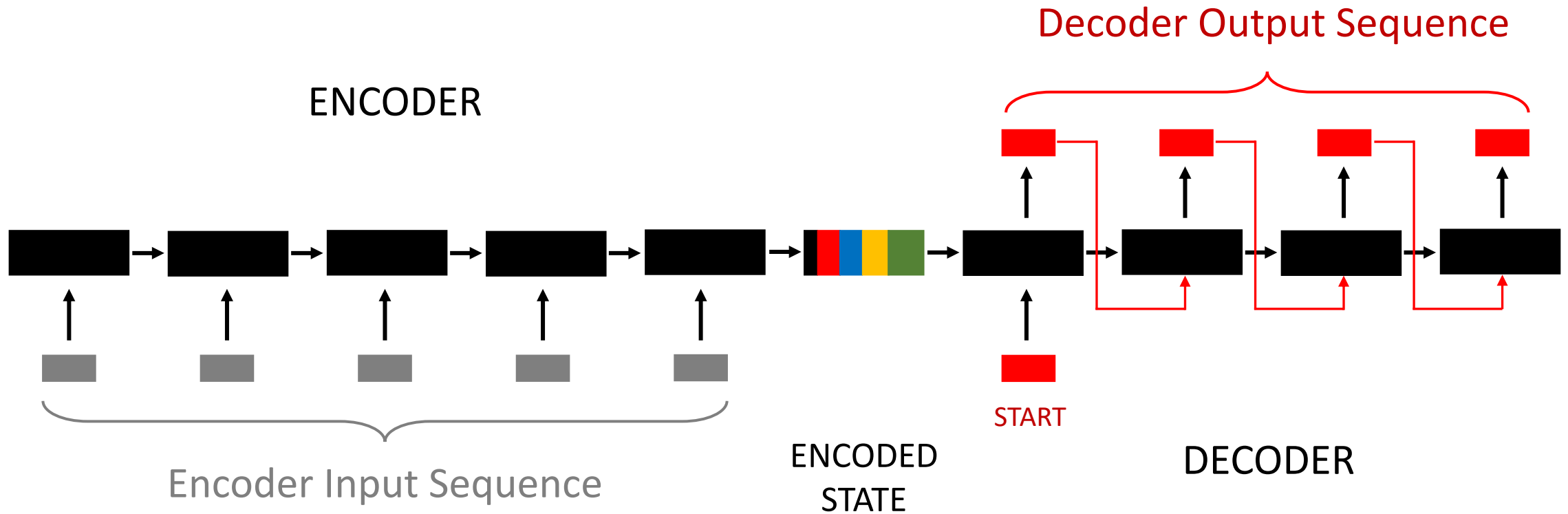
RNN Architecture



If using time-synced many-to-many →
 $len(input\ seq) == len(output\ seq)$



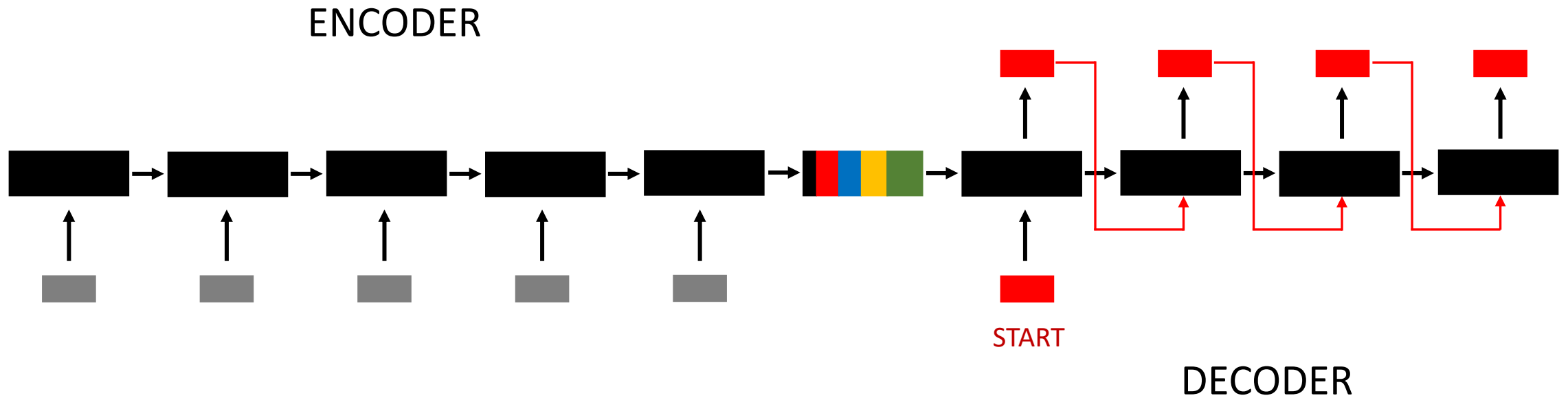
Seq2Seq



(+) Can be trained to translate input sequence to output sequence with two different lengths



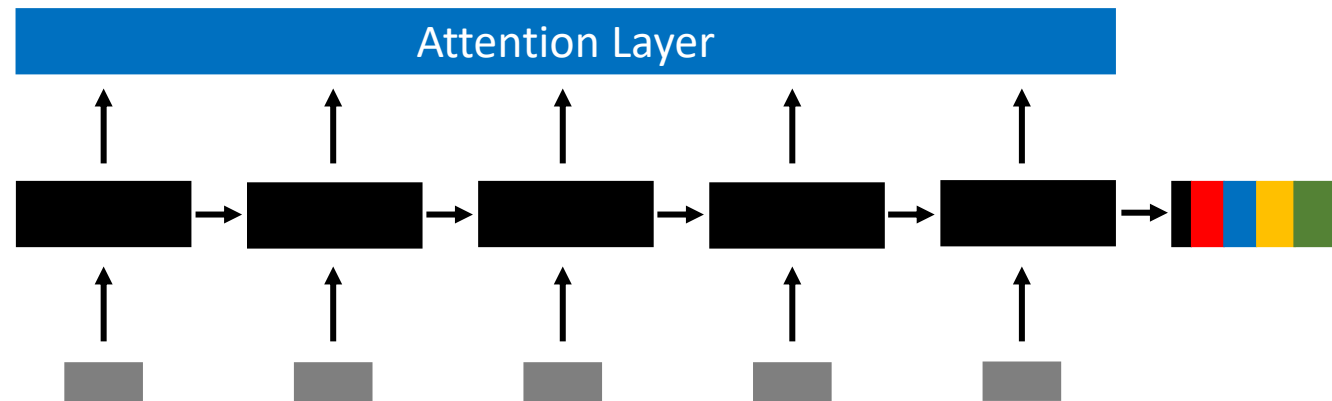
Seq2Seq



(-) Suffers from identical limitations as RNNs →
Can't process long context, Hard to parallelize

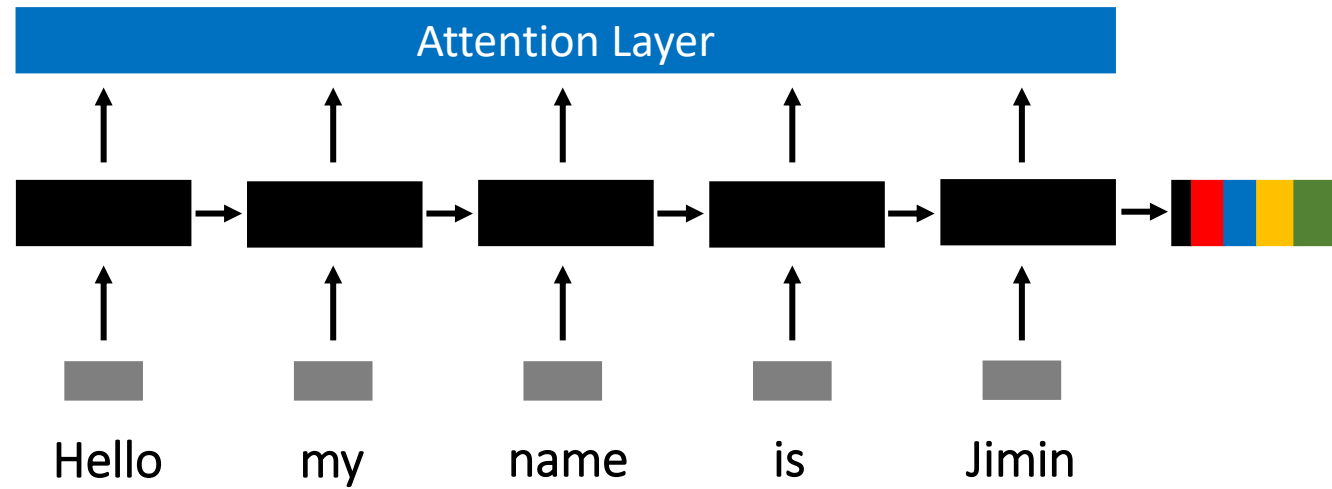


Seq2Seq with Attention



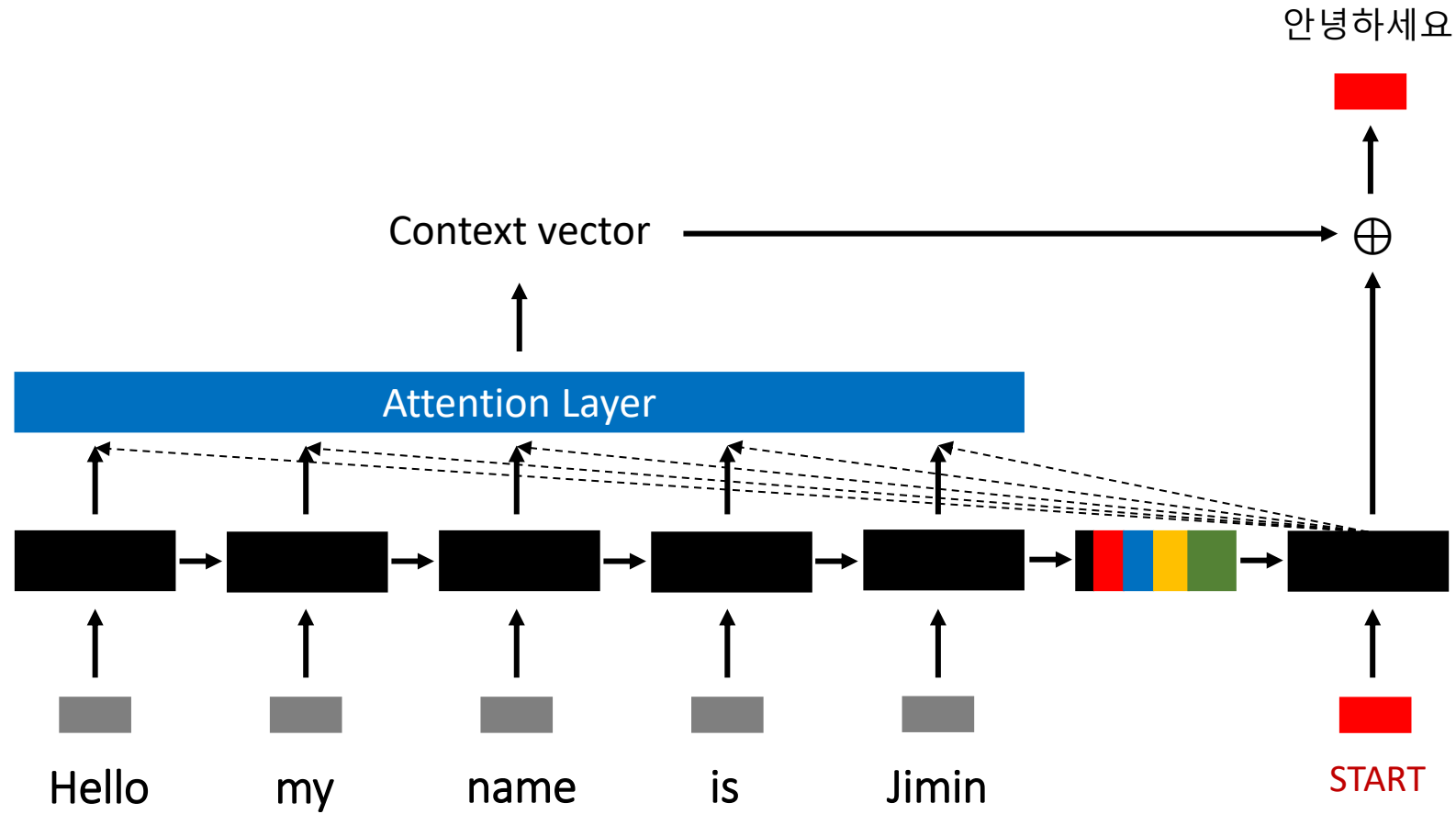


Seq2Seq with Attention



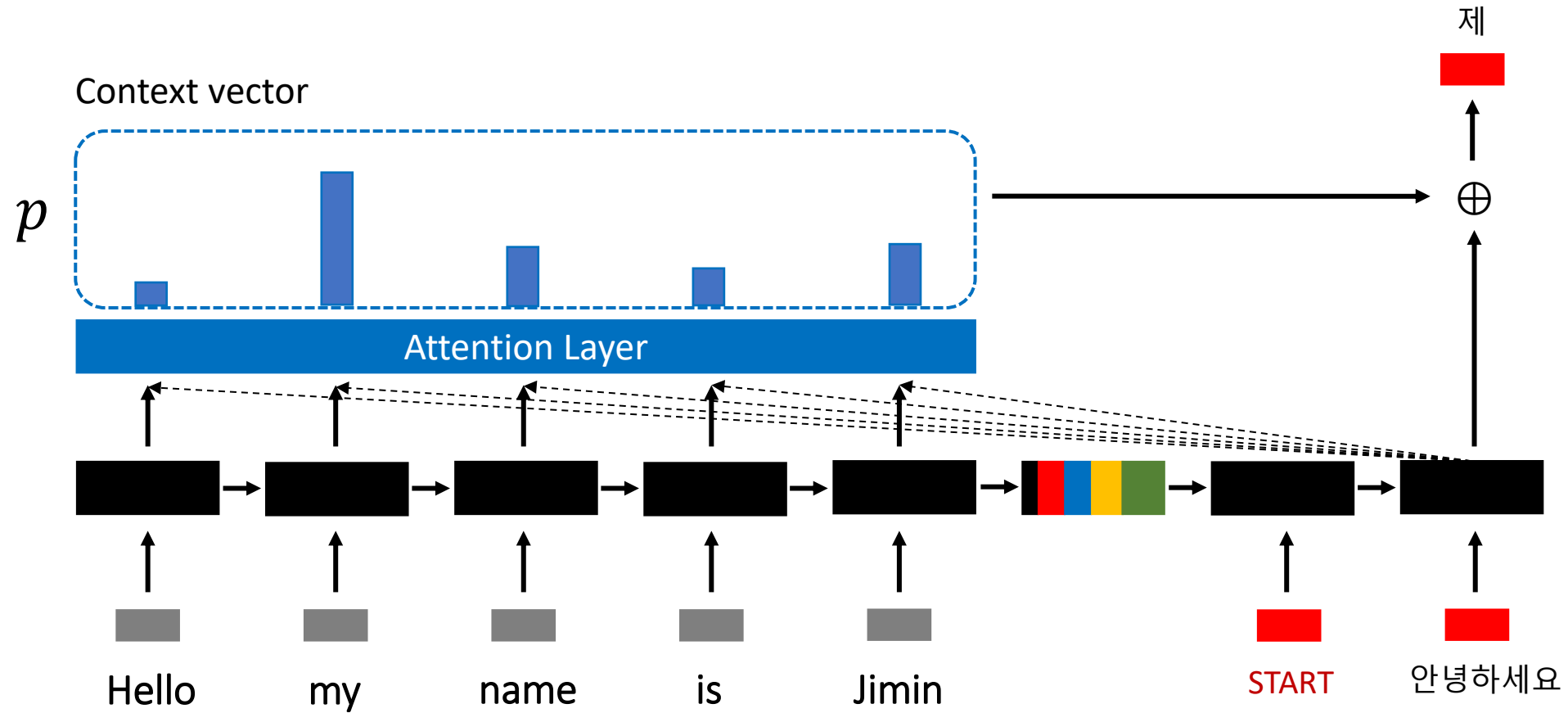


Seq2Seq with Attention



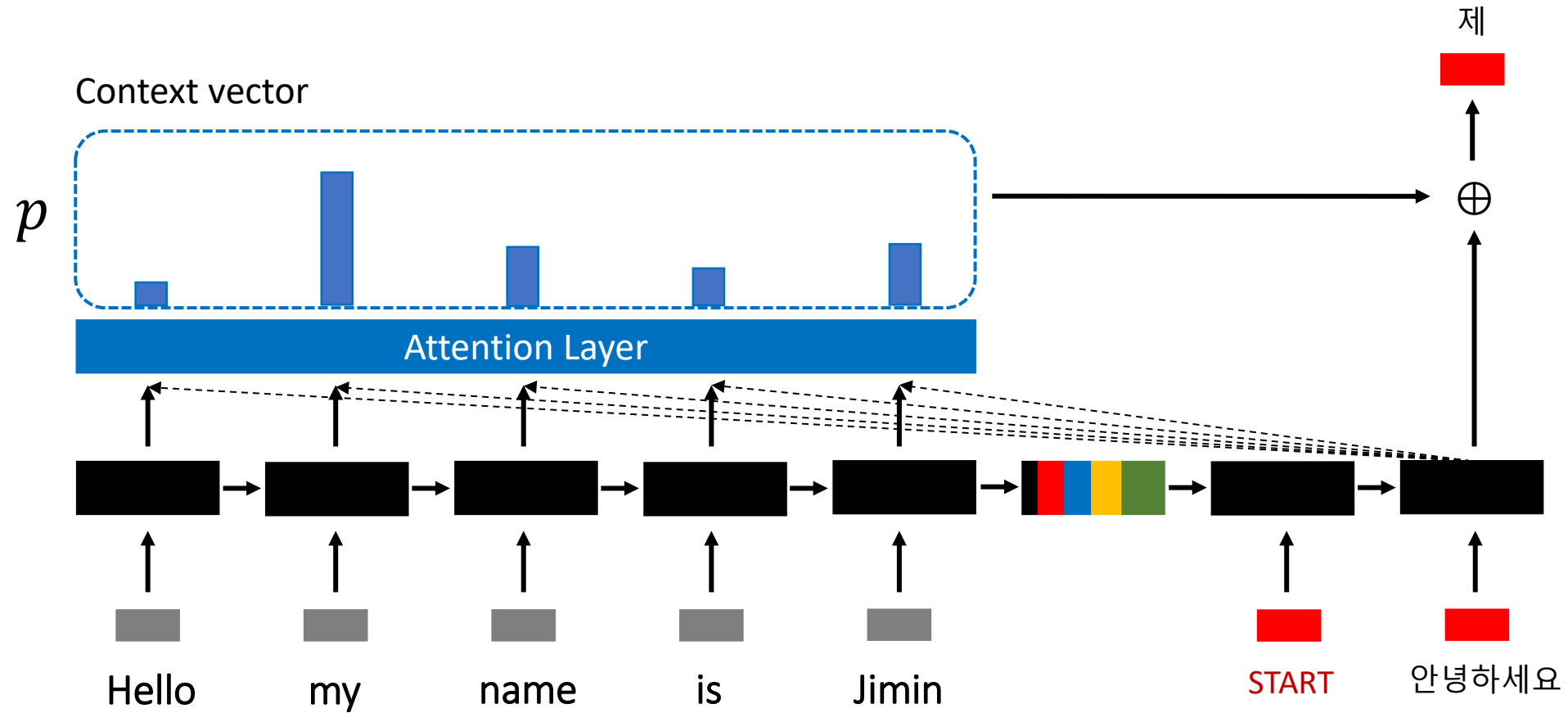


Seq2Seq with Attention





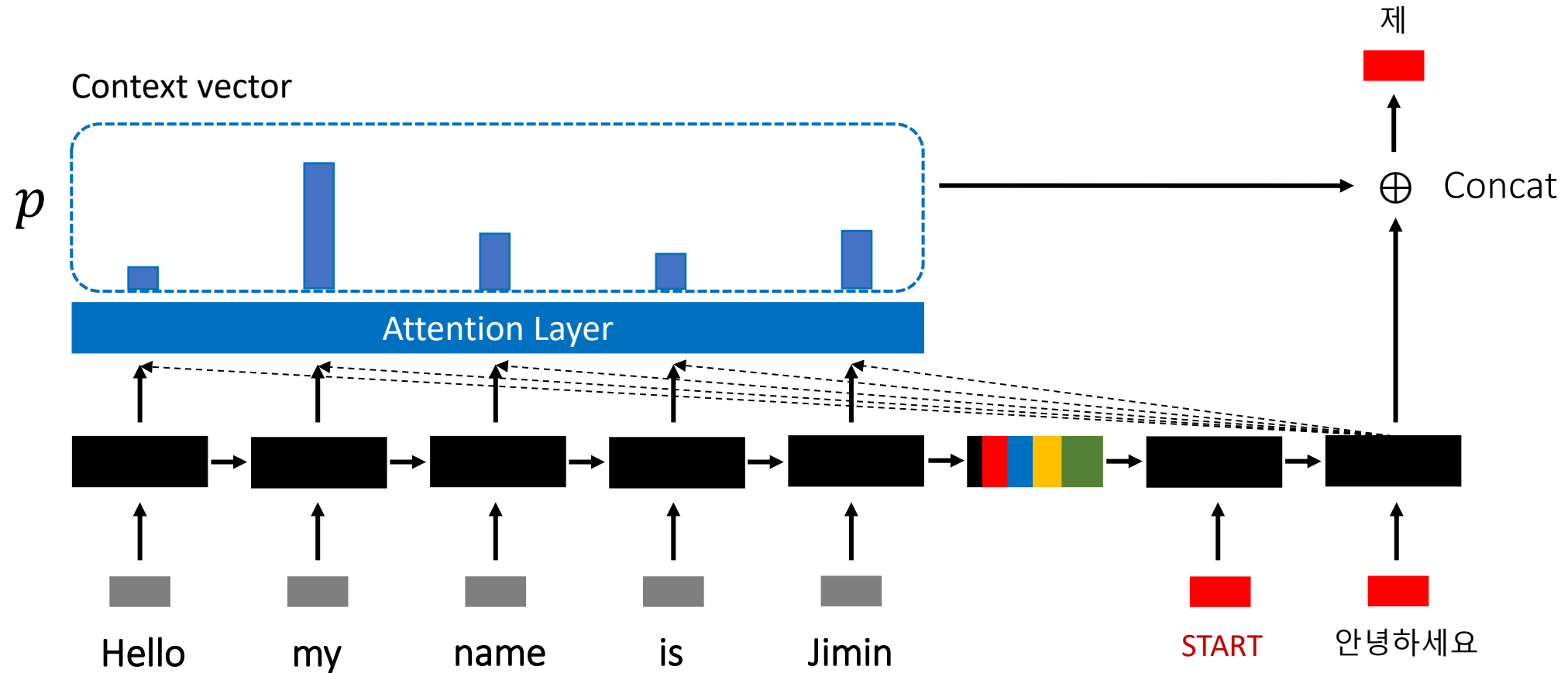
Seq2Seq with Attention



(+) Addresses long context issue



Seq2Seq with Attention

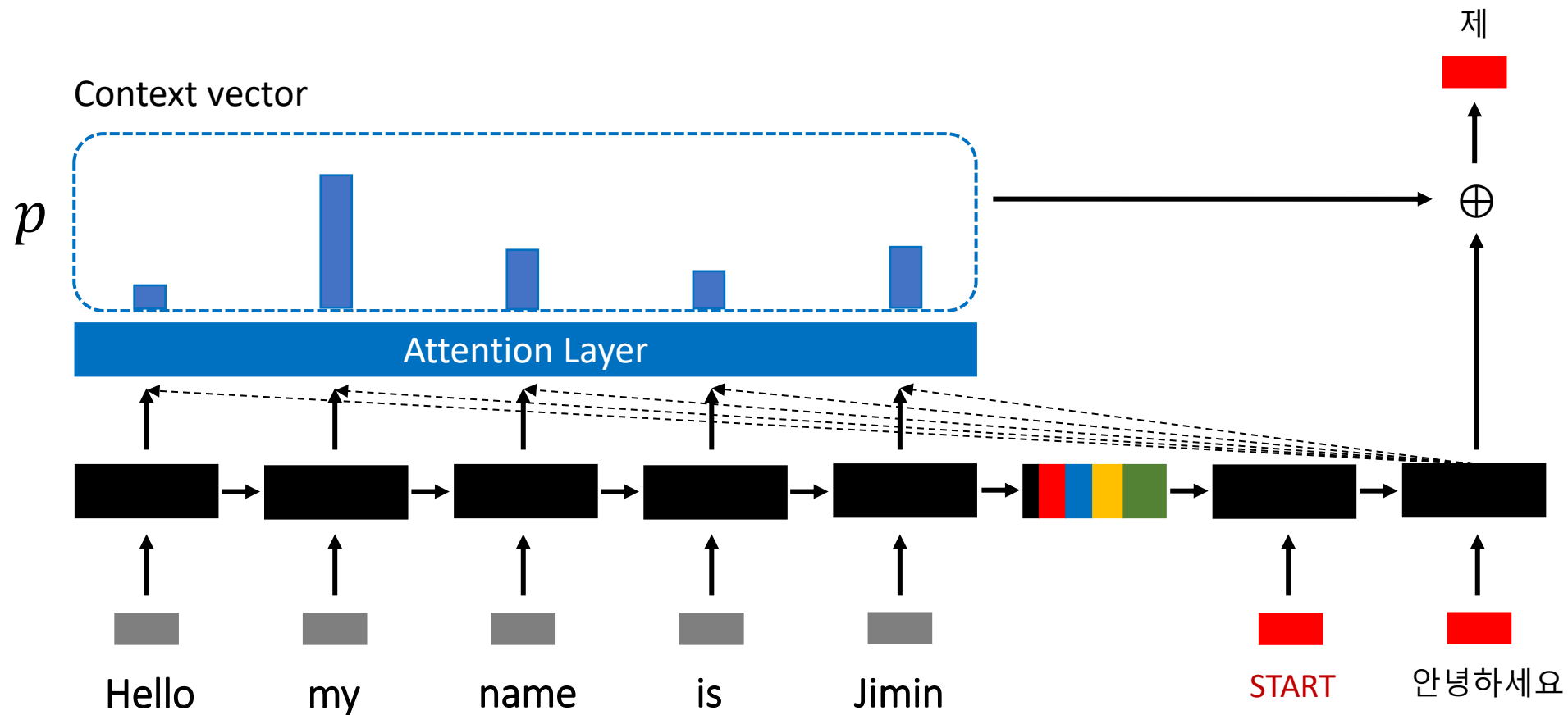


(+) Addresses long context issue

(-) Difficult to parallelize

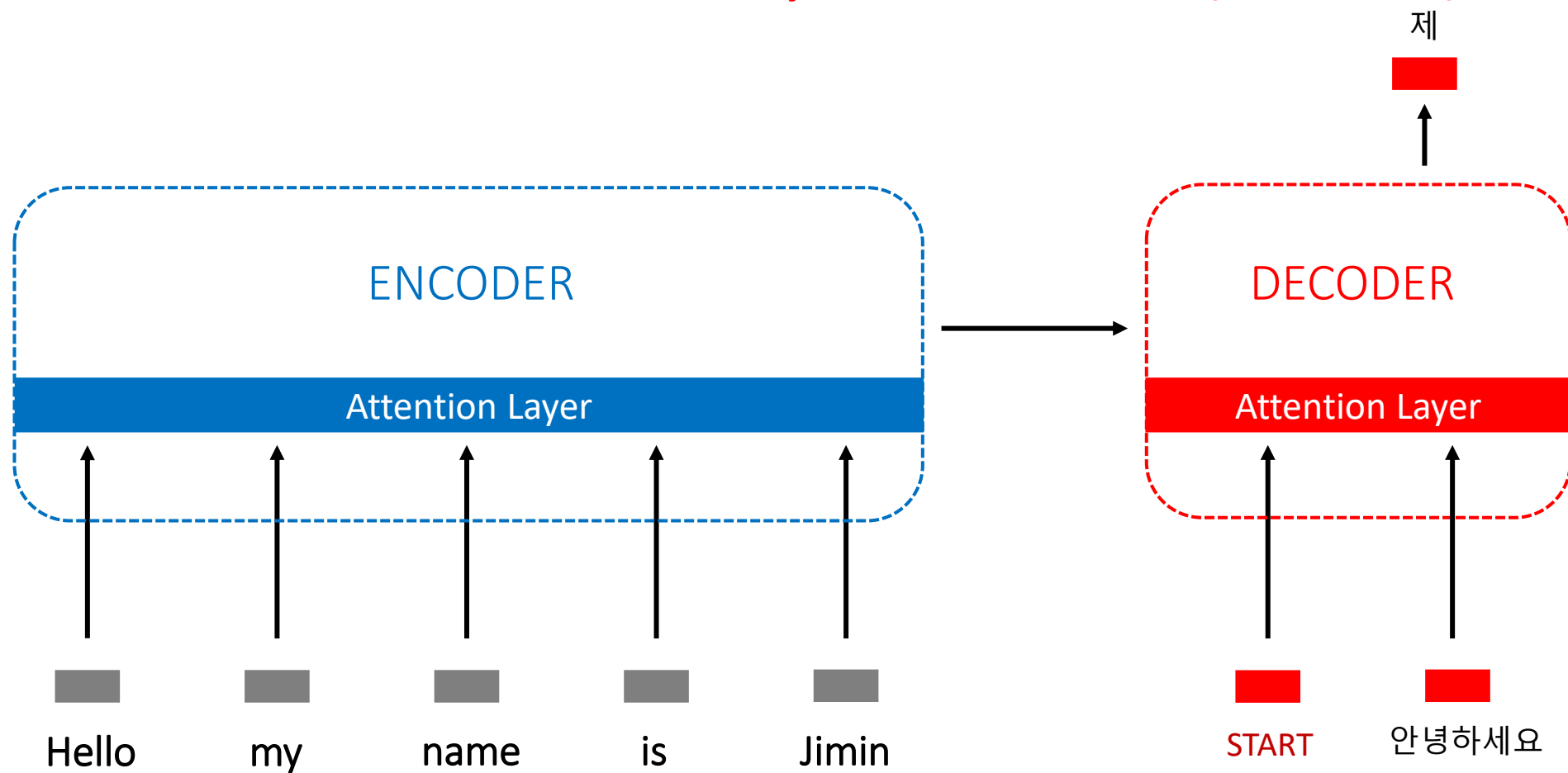


Attention is all you need (2017)



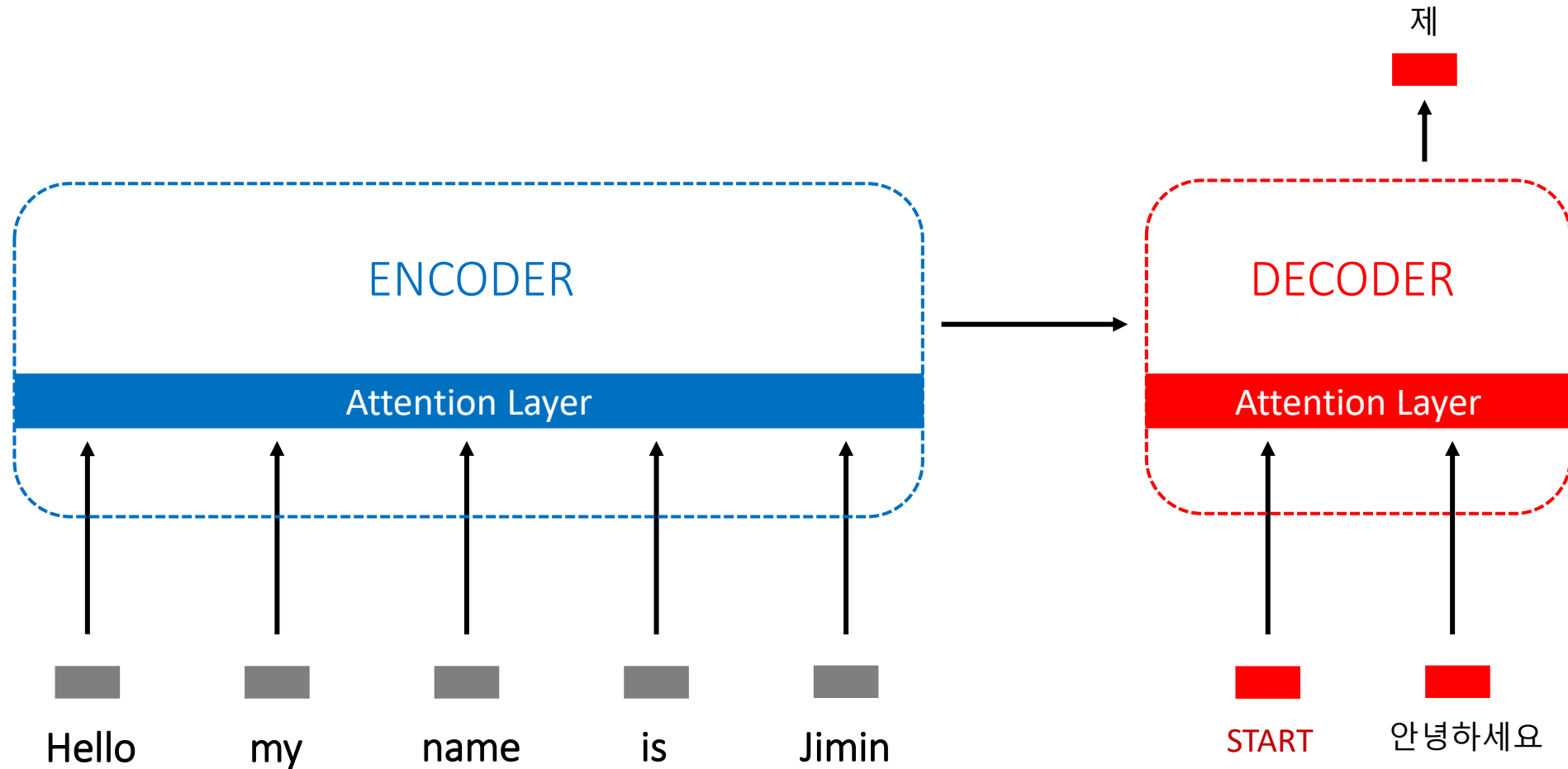


Attention is all you need (2017)





Attention is all you need (2017)



Attention without RNN is sufficient
Can utilize parallelization with GPUs



Transformer Architecture

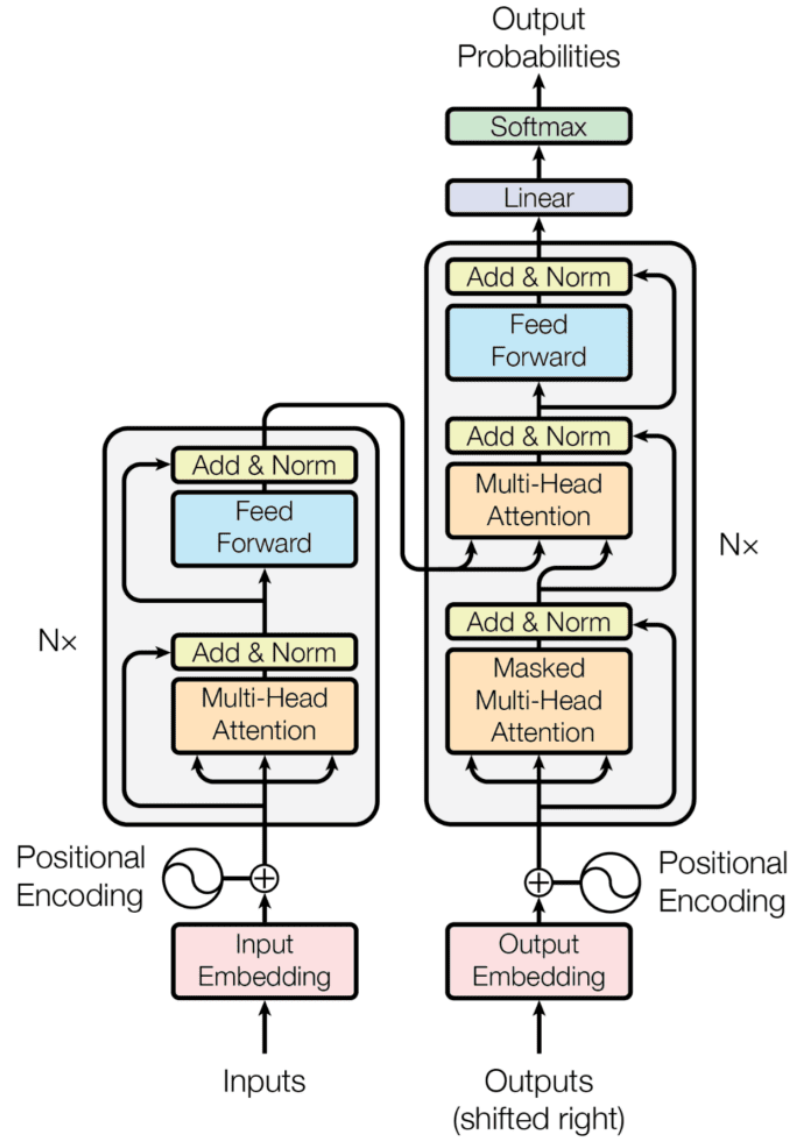
Encoder

Decoder

Transformer vs RNN

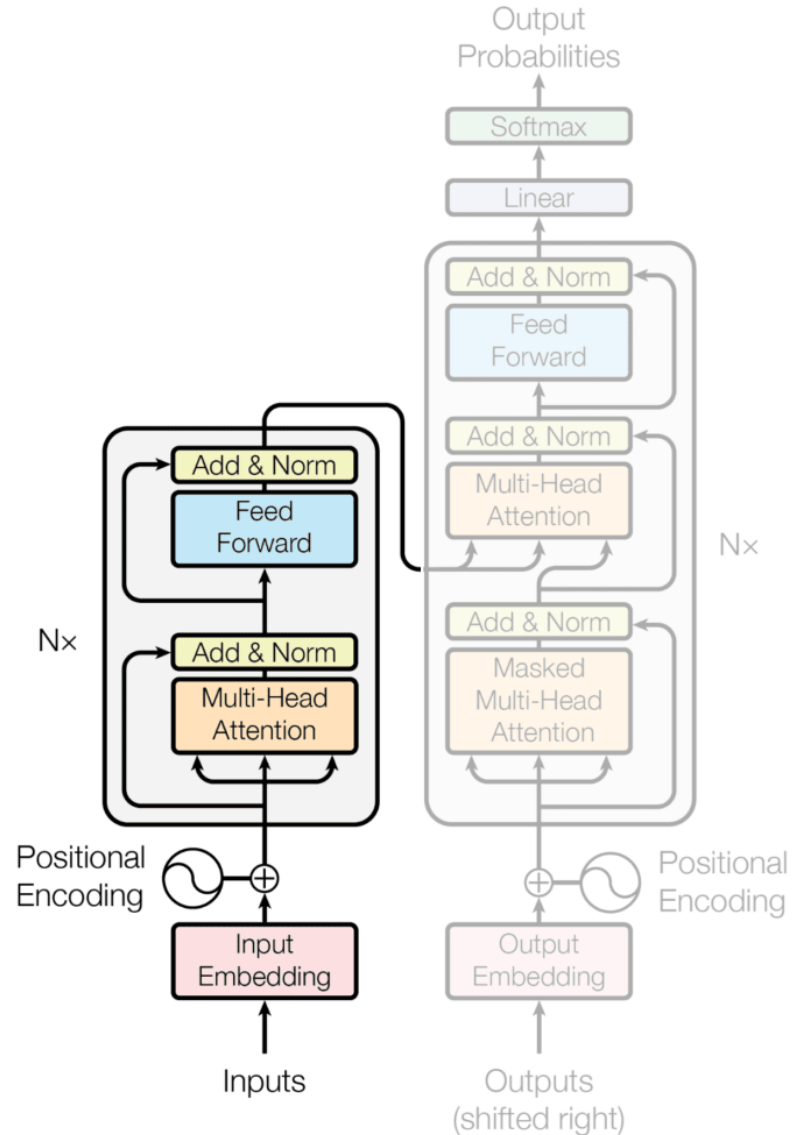


Transformer Architecture





Encoder

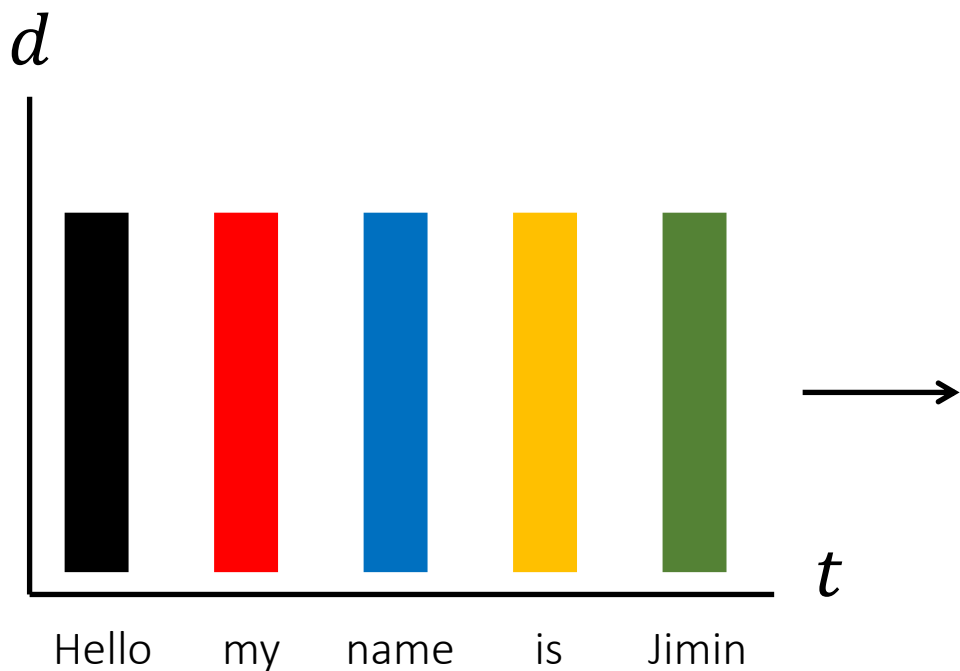


Encoder layer with

- Input embedding with positional encoding
- multi-headed self attention
- Residual connections, Layer norm & dropout

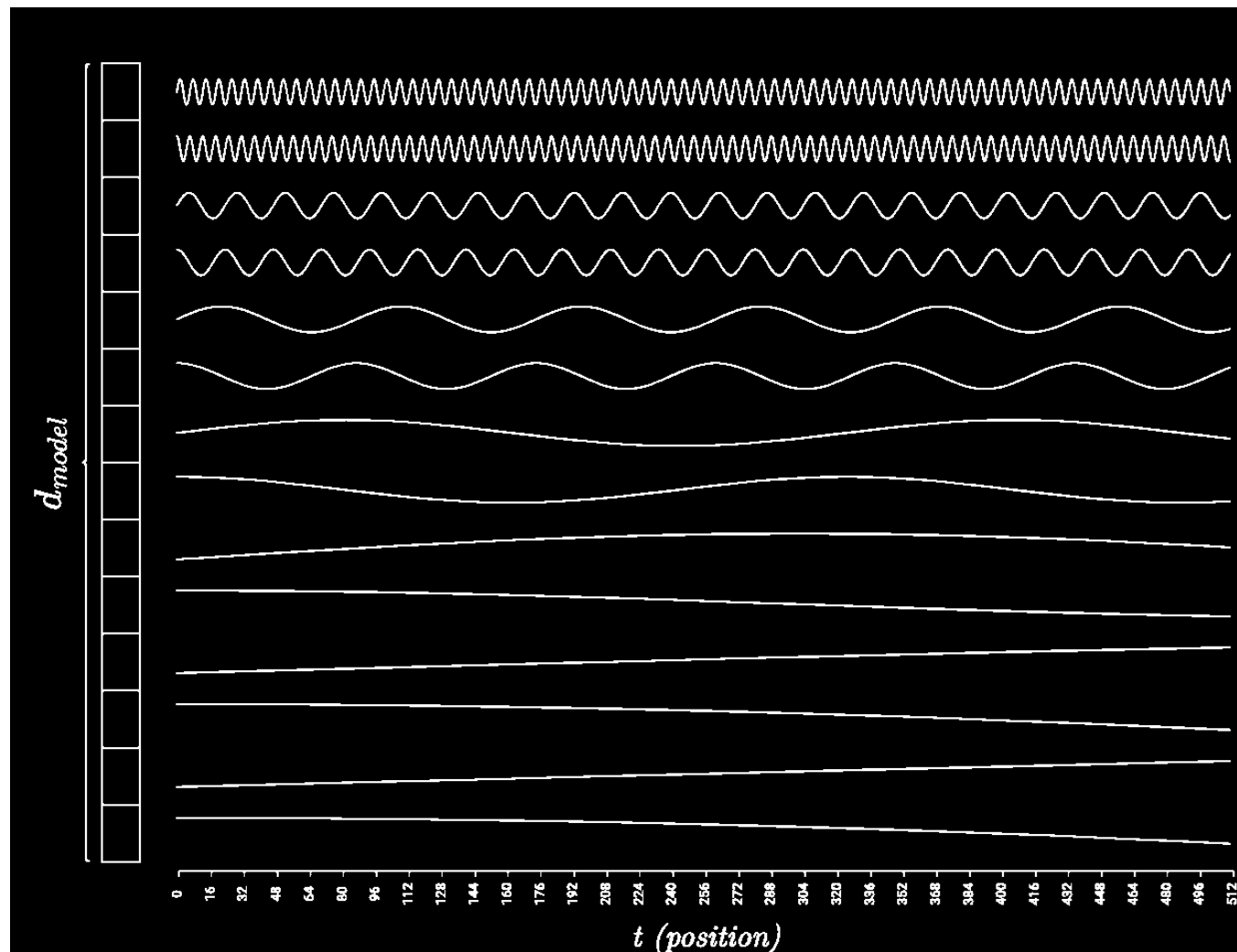


Positional Encoding



$$k \text{ is even: } \sin\left(\frac{t}{10000^{\frac{k}{d_{\text{embedding}}}}}\right)$$

$$k \text{ is odd: } \cos\left(\frac{t}{10000^{\frac{k}{d_{\text{embedding}}}}}\right)$$





Self-attention layer

Overview

Key, Query, Value retrieval process

Multi-headed attention



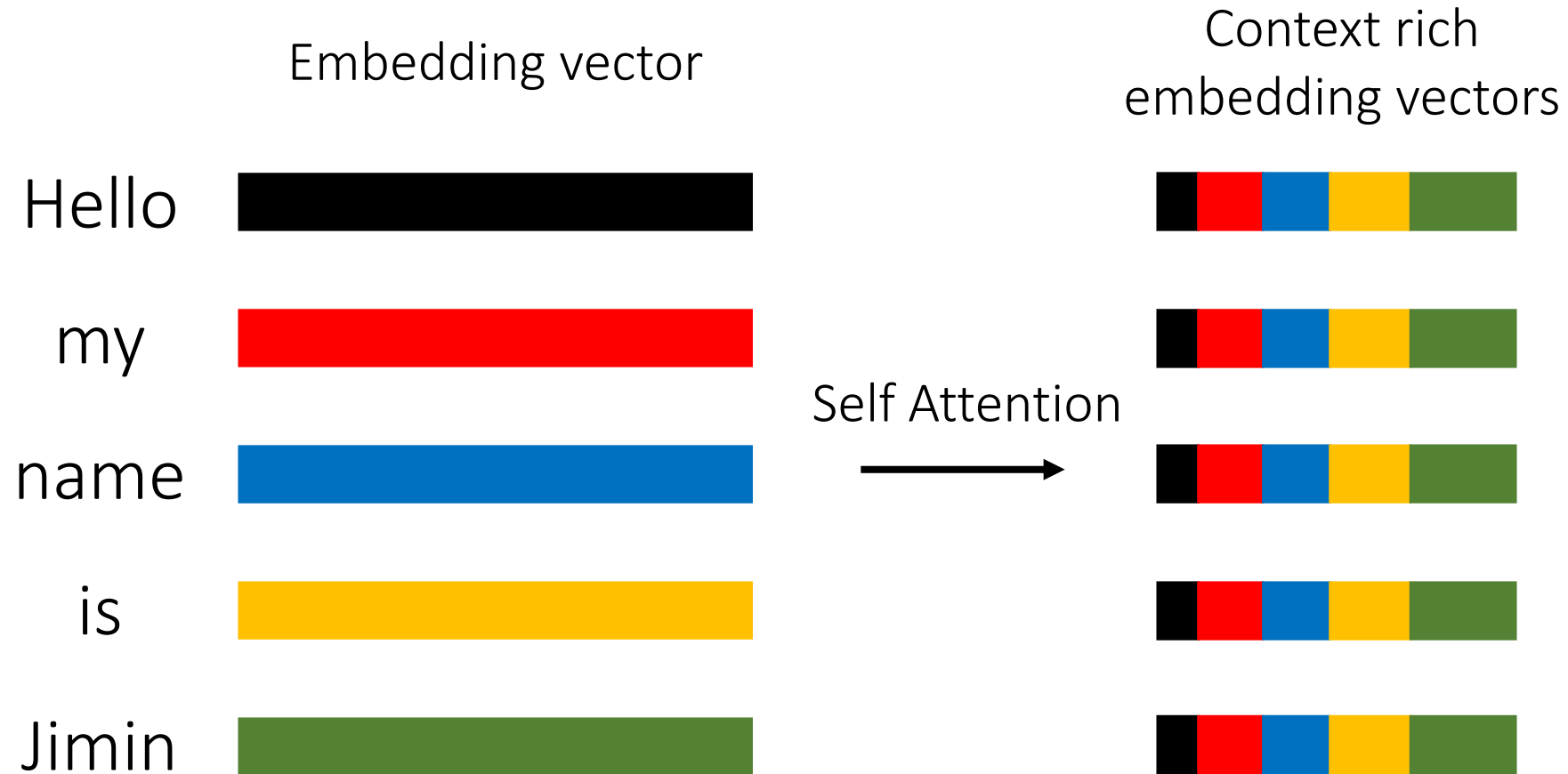
Overview of self-attention layer

Embedding vector

Hello	
my	
name	
is	
Jimin	

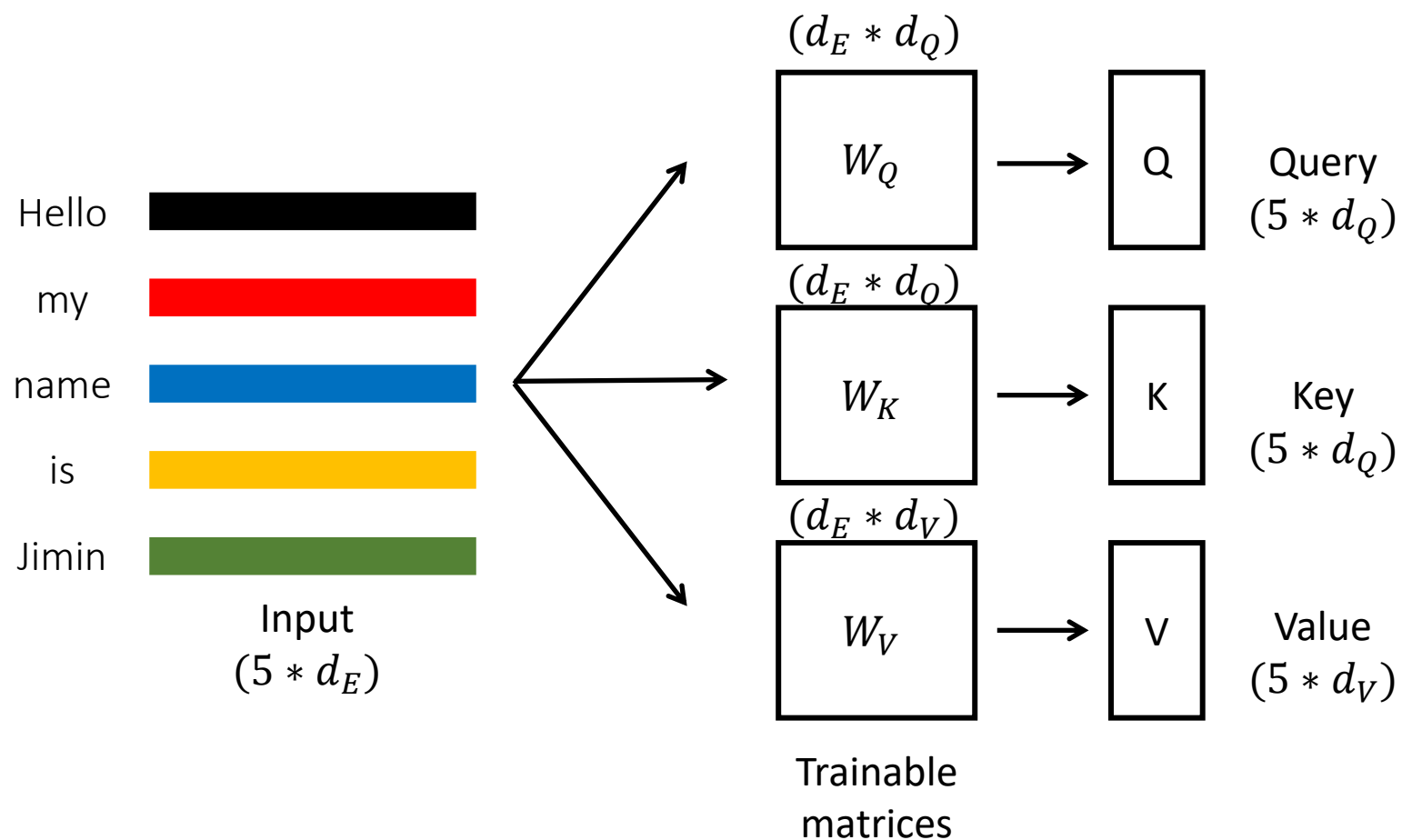


Overview of self-attention layer



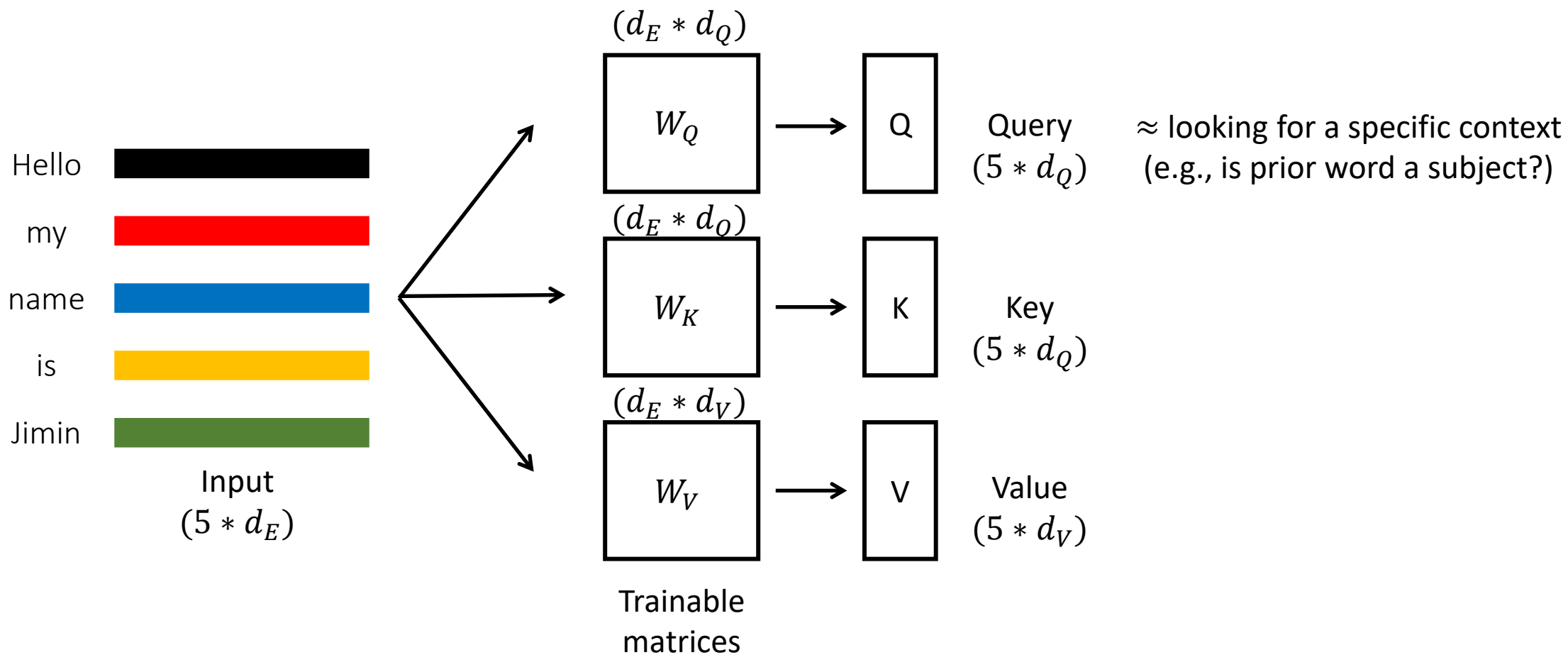


Key, Query, Value retrieval



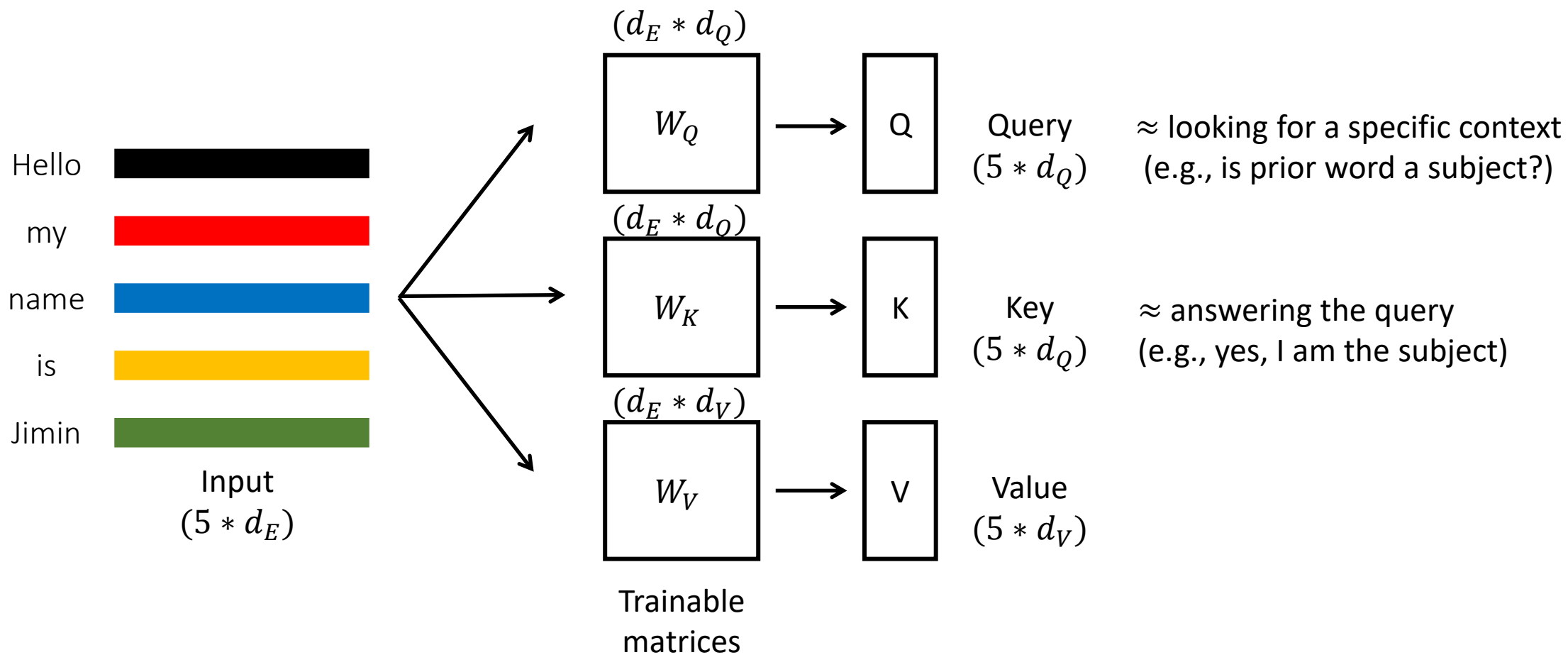


Key, Query, Value retrieval



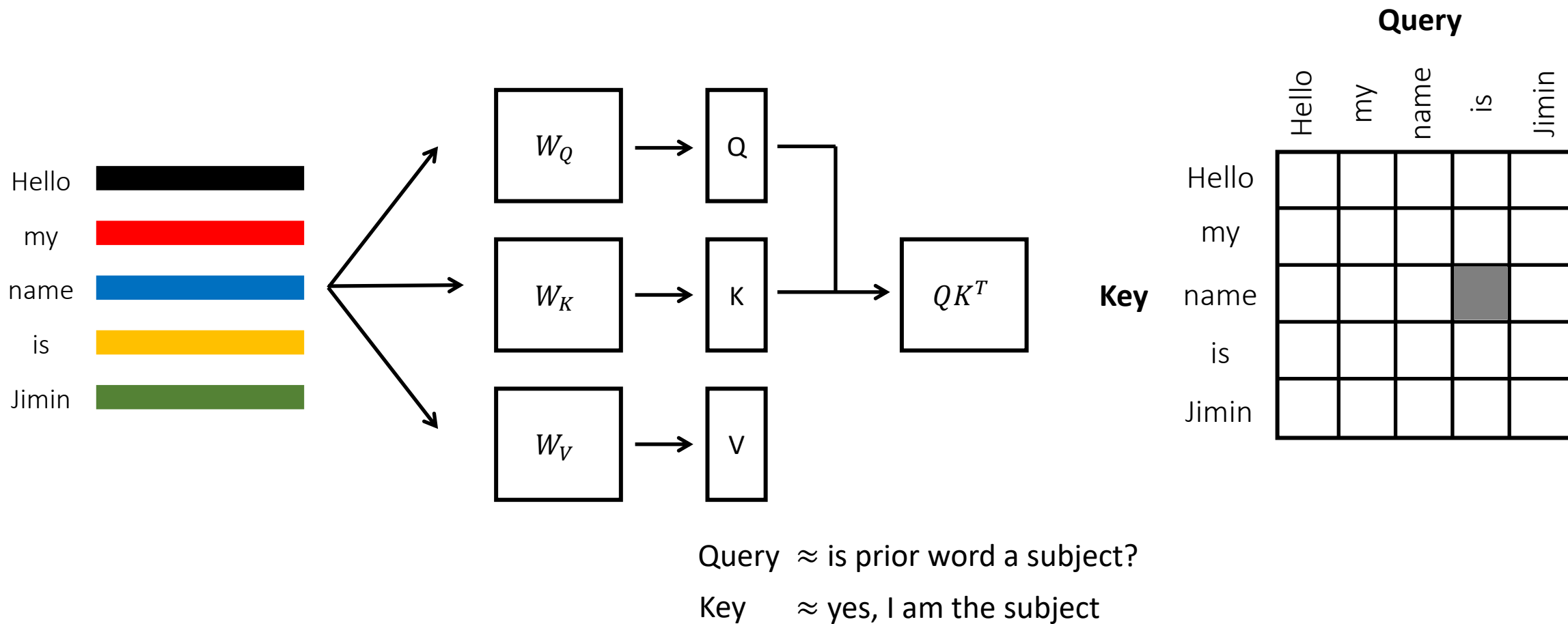


Key, Query, Value retrieval



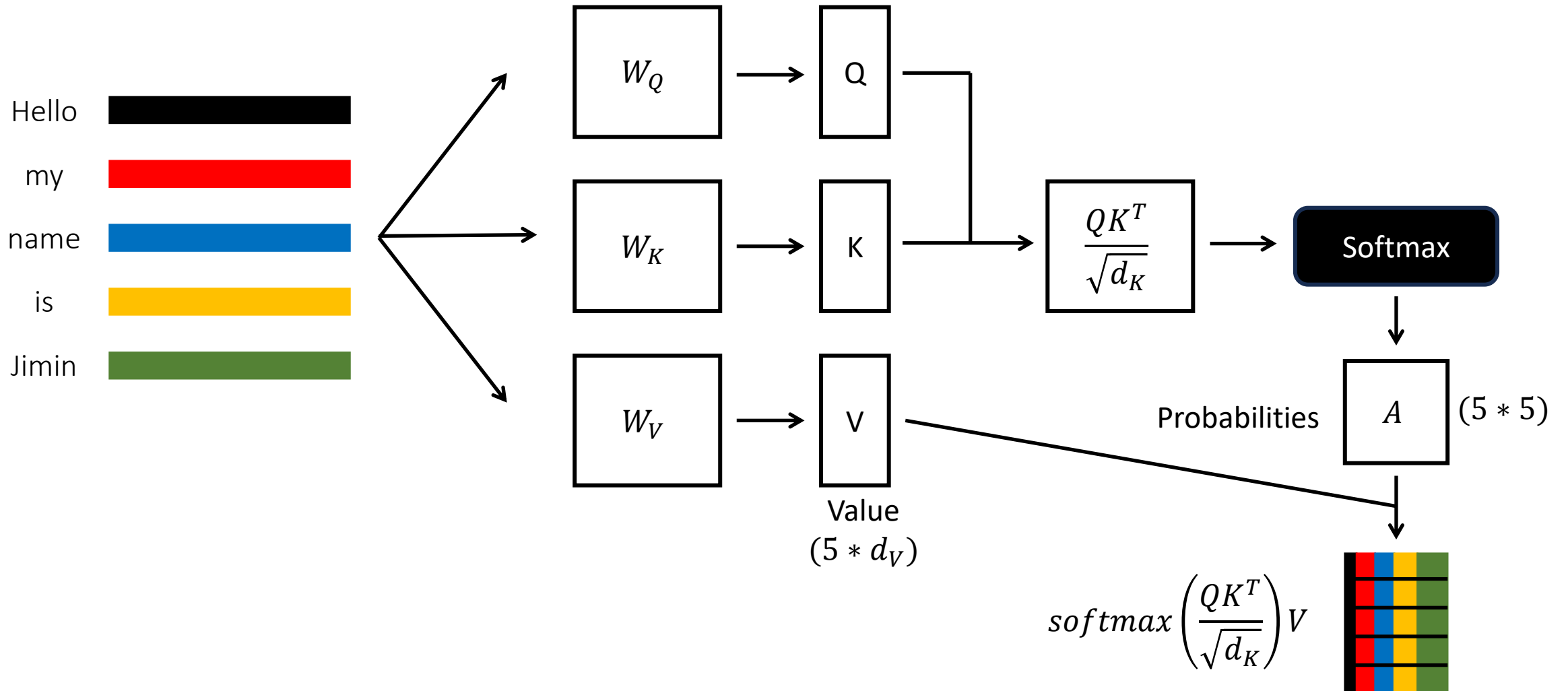


Key, Query, Value retrieval



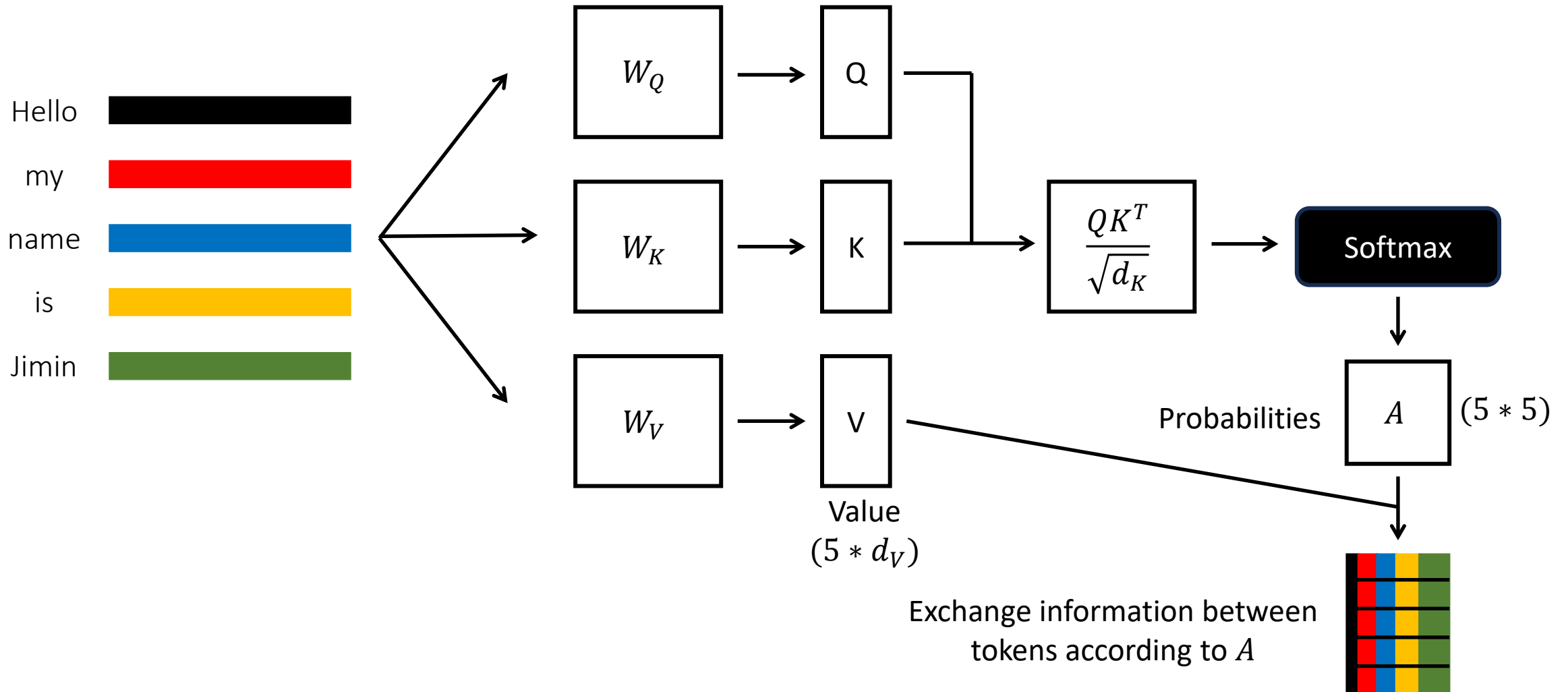


Key, Query, Value retrieval



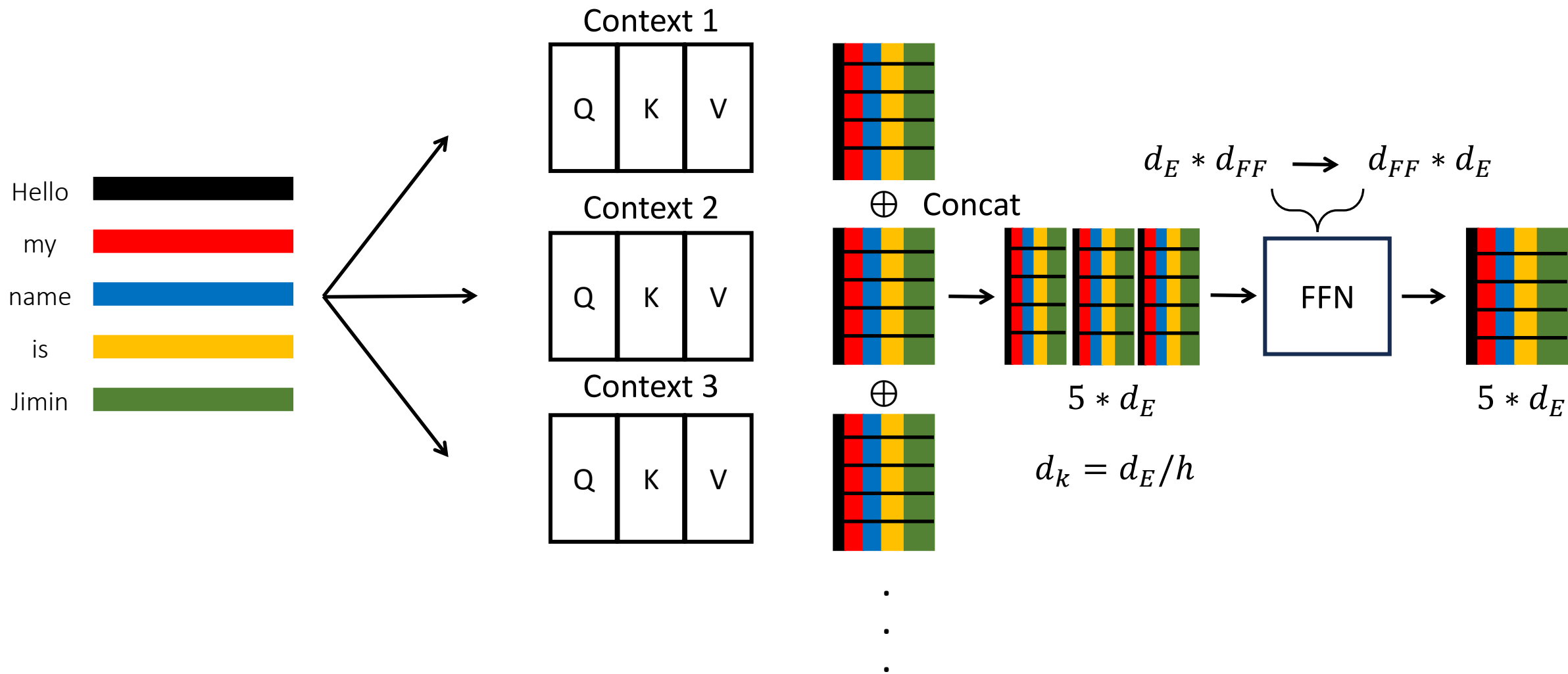


Key, Query, Value retrieval



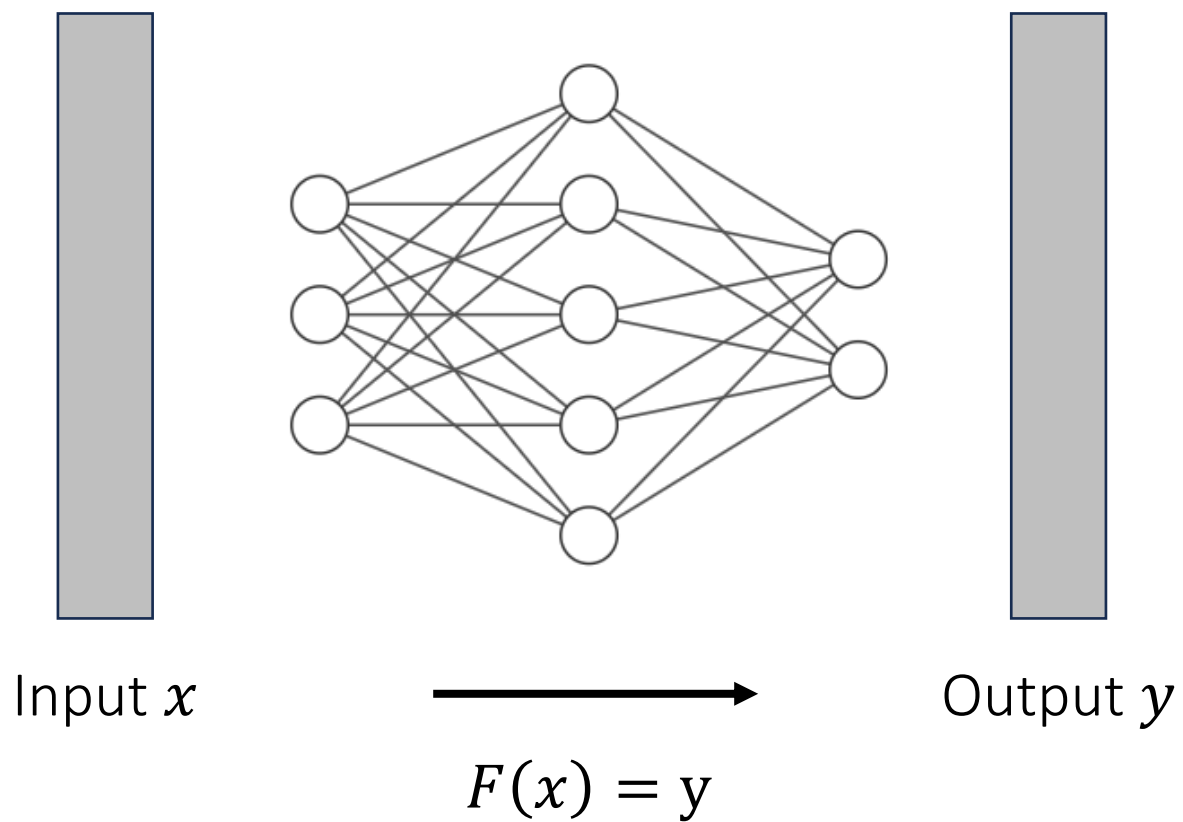


Multi-headed attention



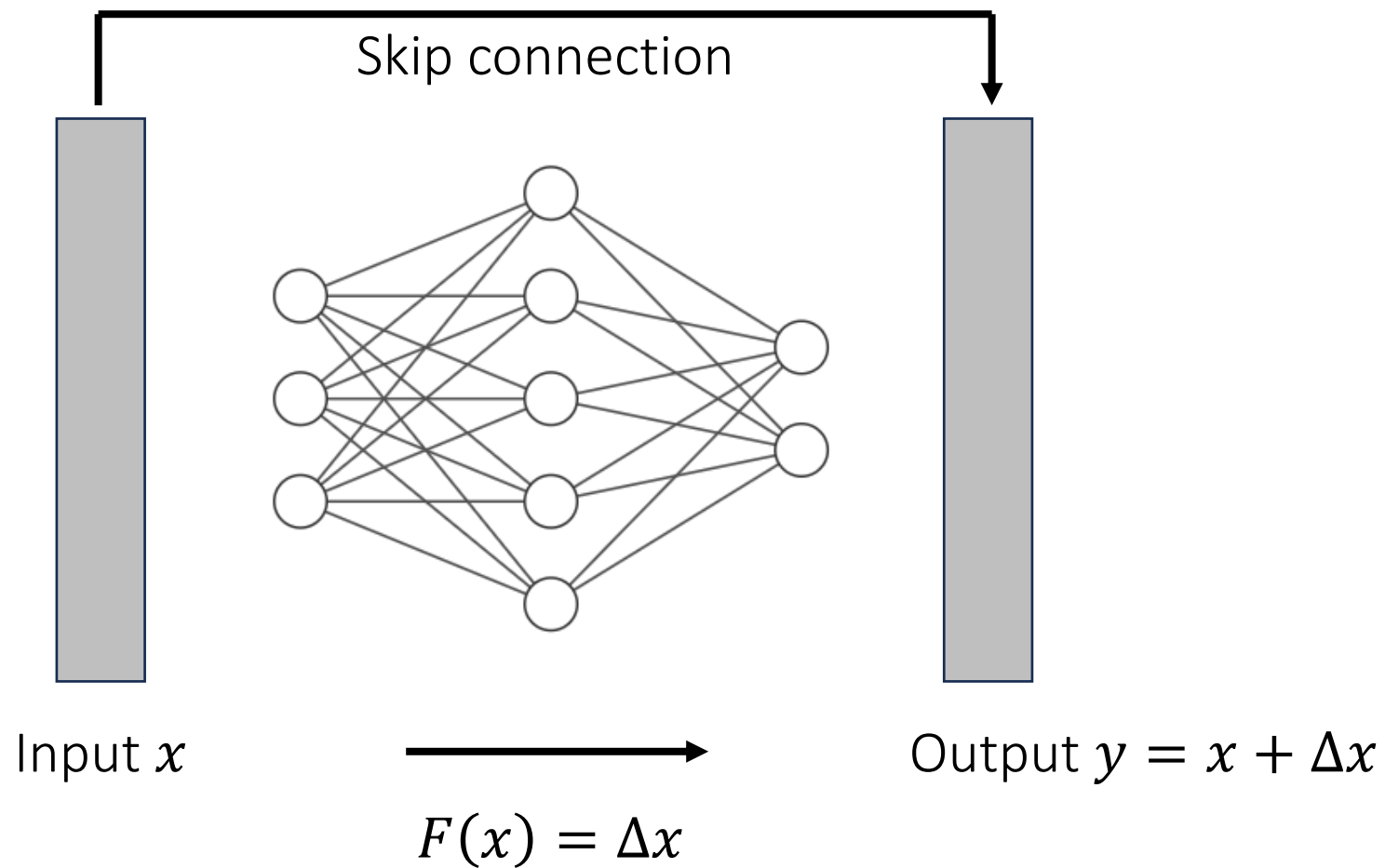


Residual connections



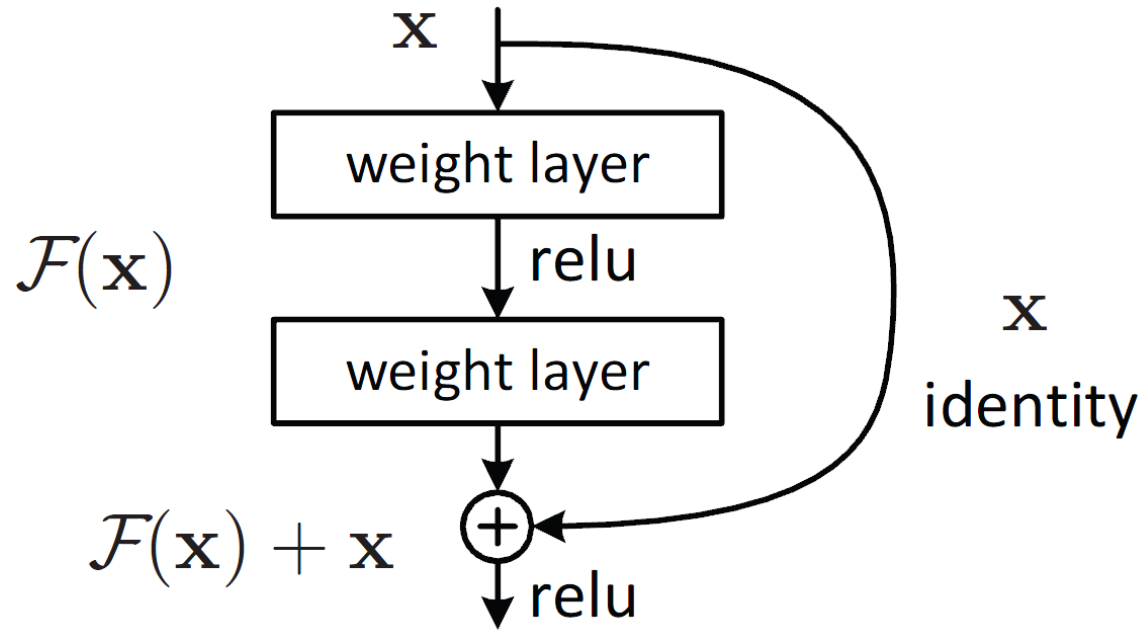


Residual connections





Residual blocks

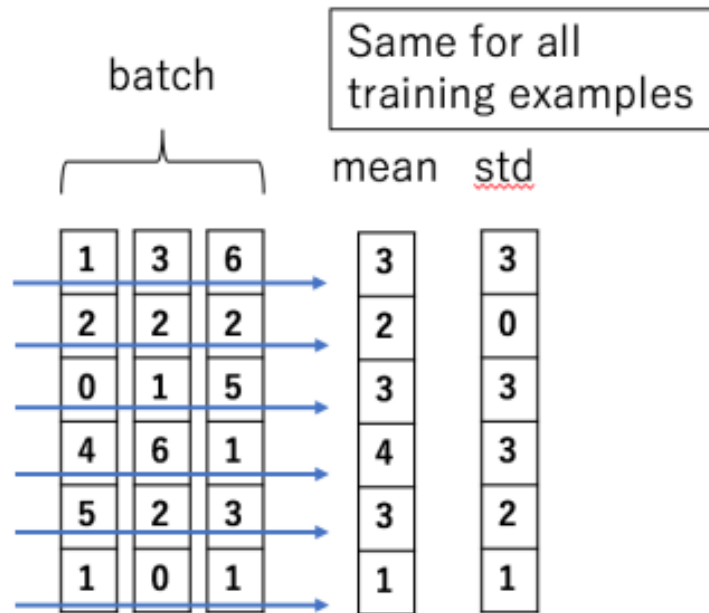


- Enable training deeper networks
- Prevents vanishing gradients
- Reduces overfitting

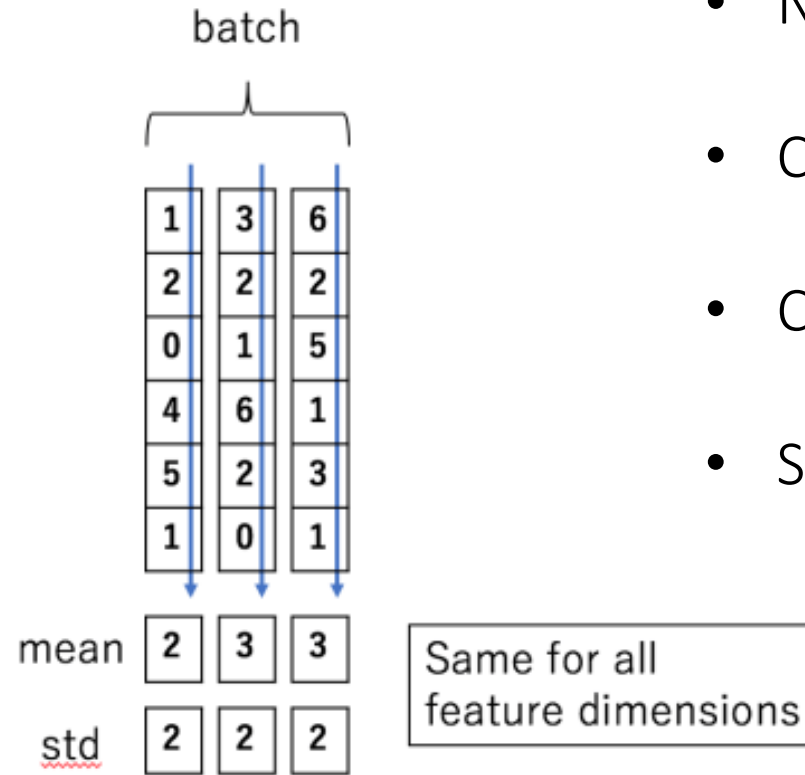


Layer Normalization

Batch Normalization



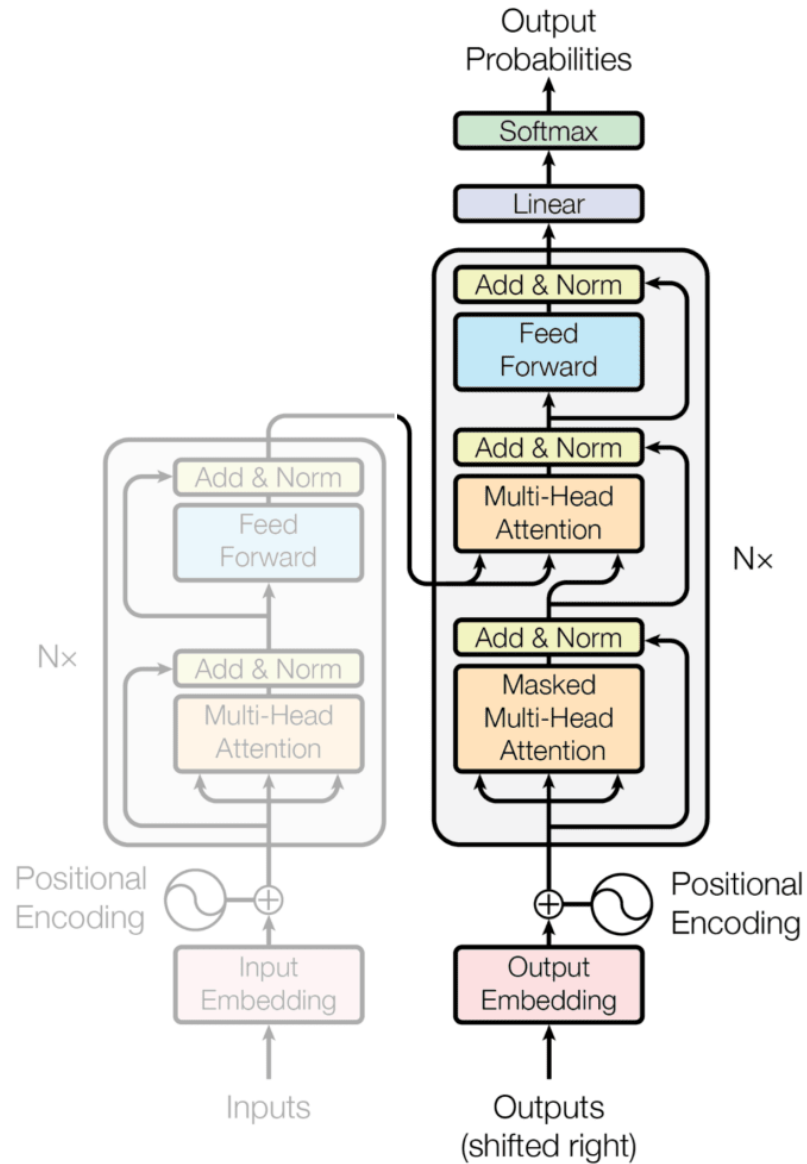
Layer Normalization



- Normalizes layer inputs
- Can stabilize training
- Can improve generalization
- Speeds up training



Decoder

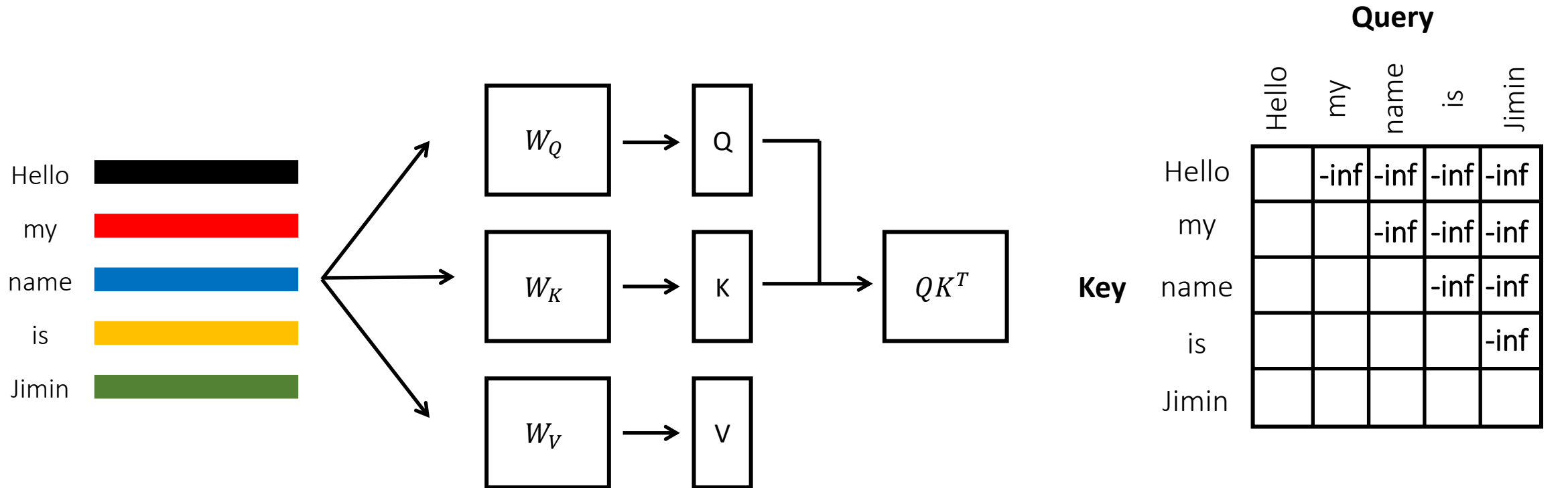


Decoder layer with

- Masked multi-headed self attention
- Multiheaded cross attention
 - Inputs $\rightarrow$ Key, Query
 - Outputs $\rightarrow$ Value



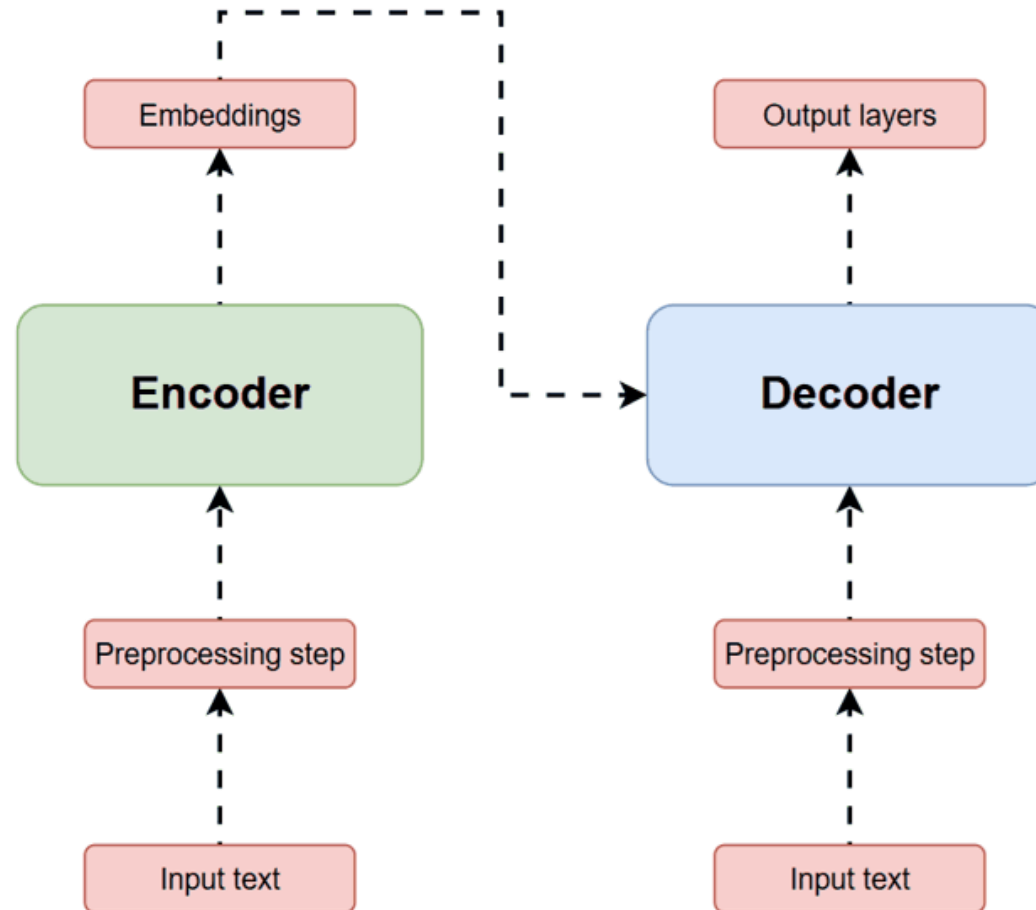
Masked Multi-headed Attention



- Can only attend to present and past
- Generation is strictly **causal**



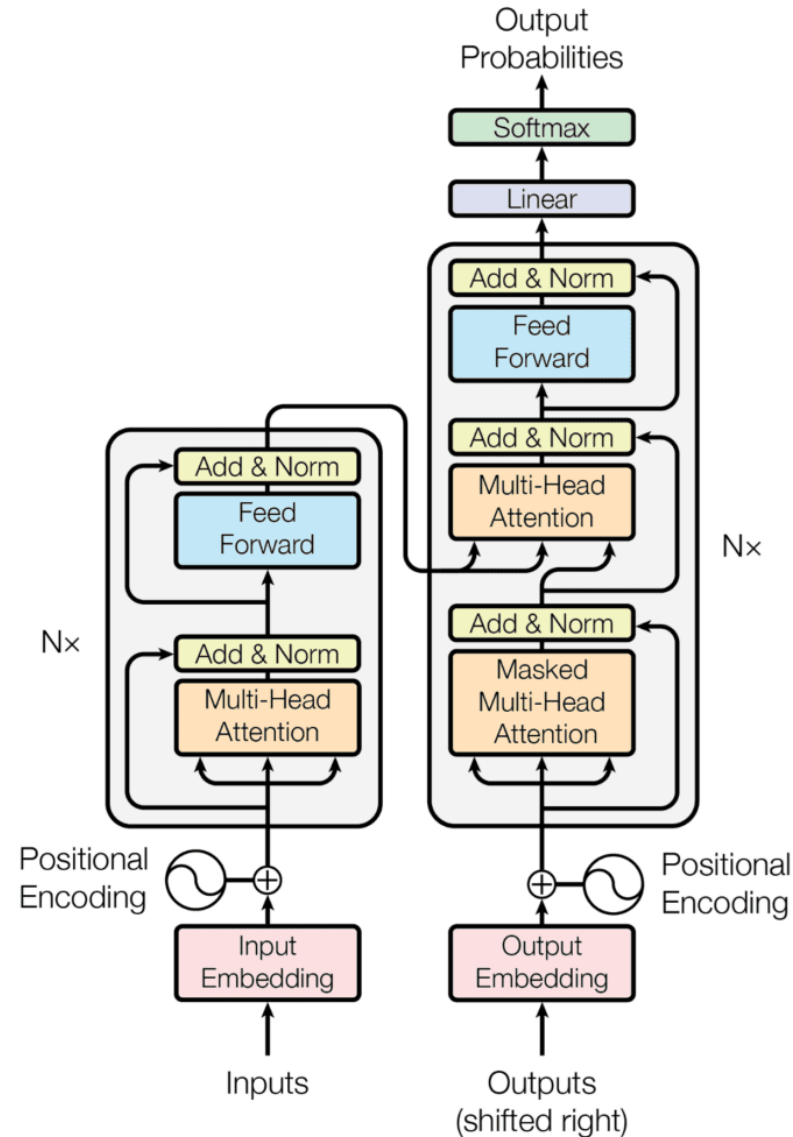
Encoder and Decoder in action



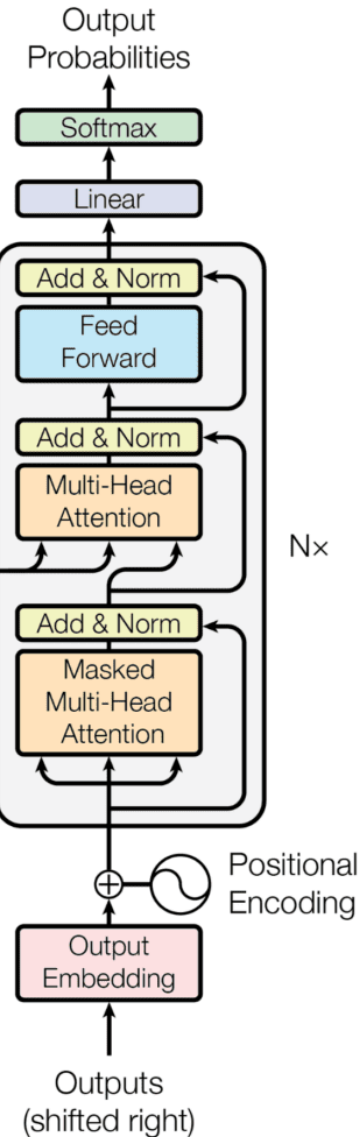


Encoder vs Decoder

BERT
(Encoder only)



GPT
(Decoder only)



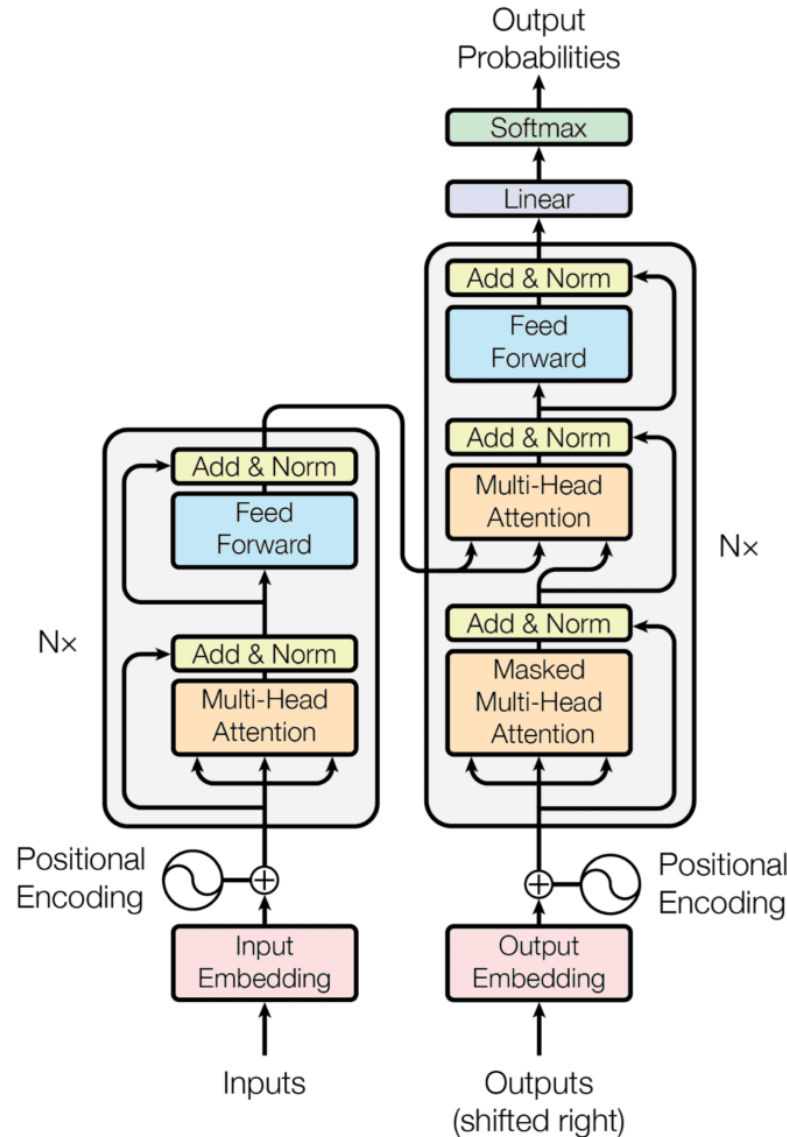


Encoder vs Decoder

BERT (Encoder only)

Bi-directional

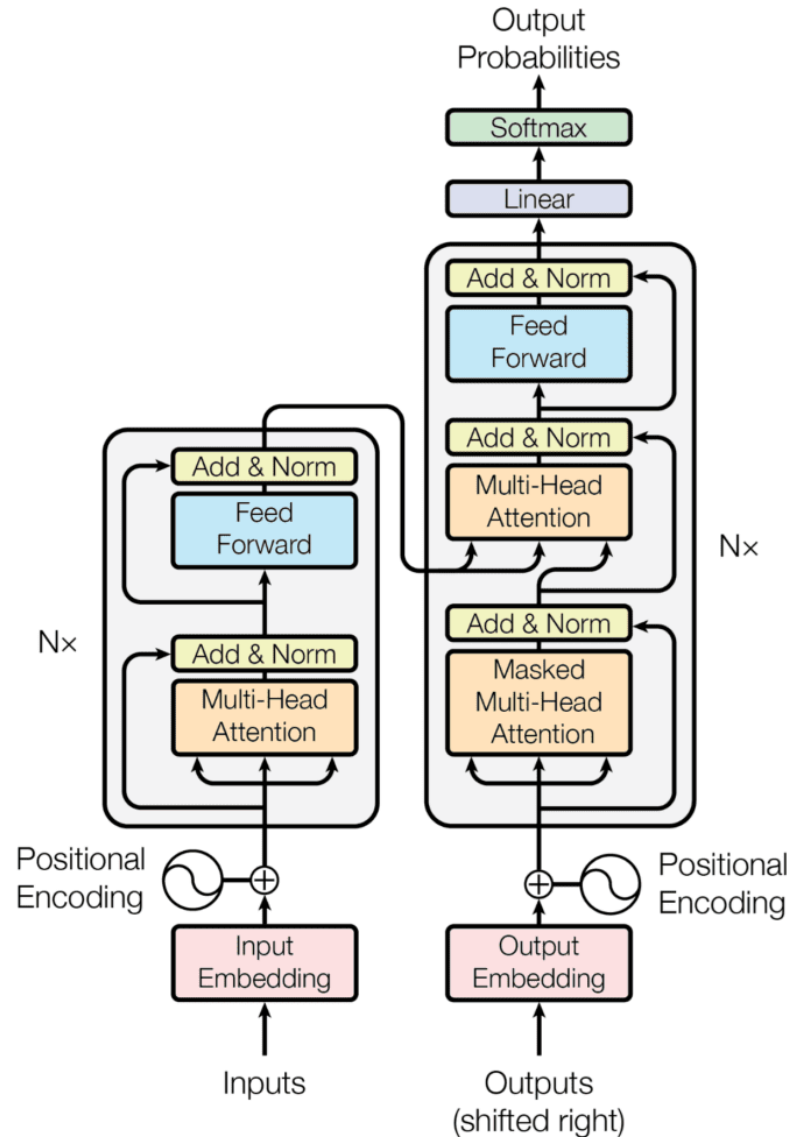
Better for understanding text
e.g., sentiment analysis



GPT (Decoder only)

Uni-directional

Better for generating text
e.g., ChatGPT





Transformer vs RNN

Transformers

RNNs

Sequential

No

Yes

Parallel computation

Yes

No

Long-term dependencies

Yes

Kind of

Scalability

Yes

Problematic

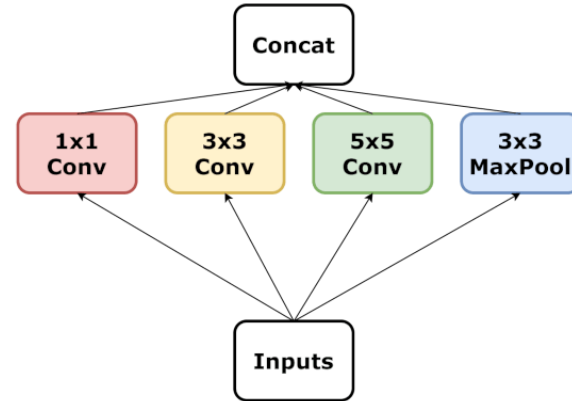
Fine tuning

Yes

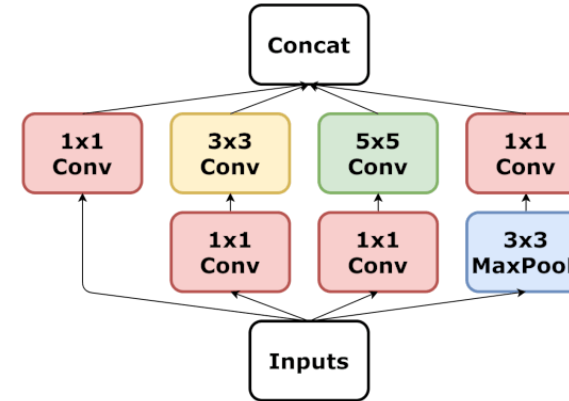
Difficult



Next episode in EE P 596...



Inception Module



Inception Module with Dimension Reduction

Advanced CNN Architectures

